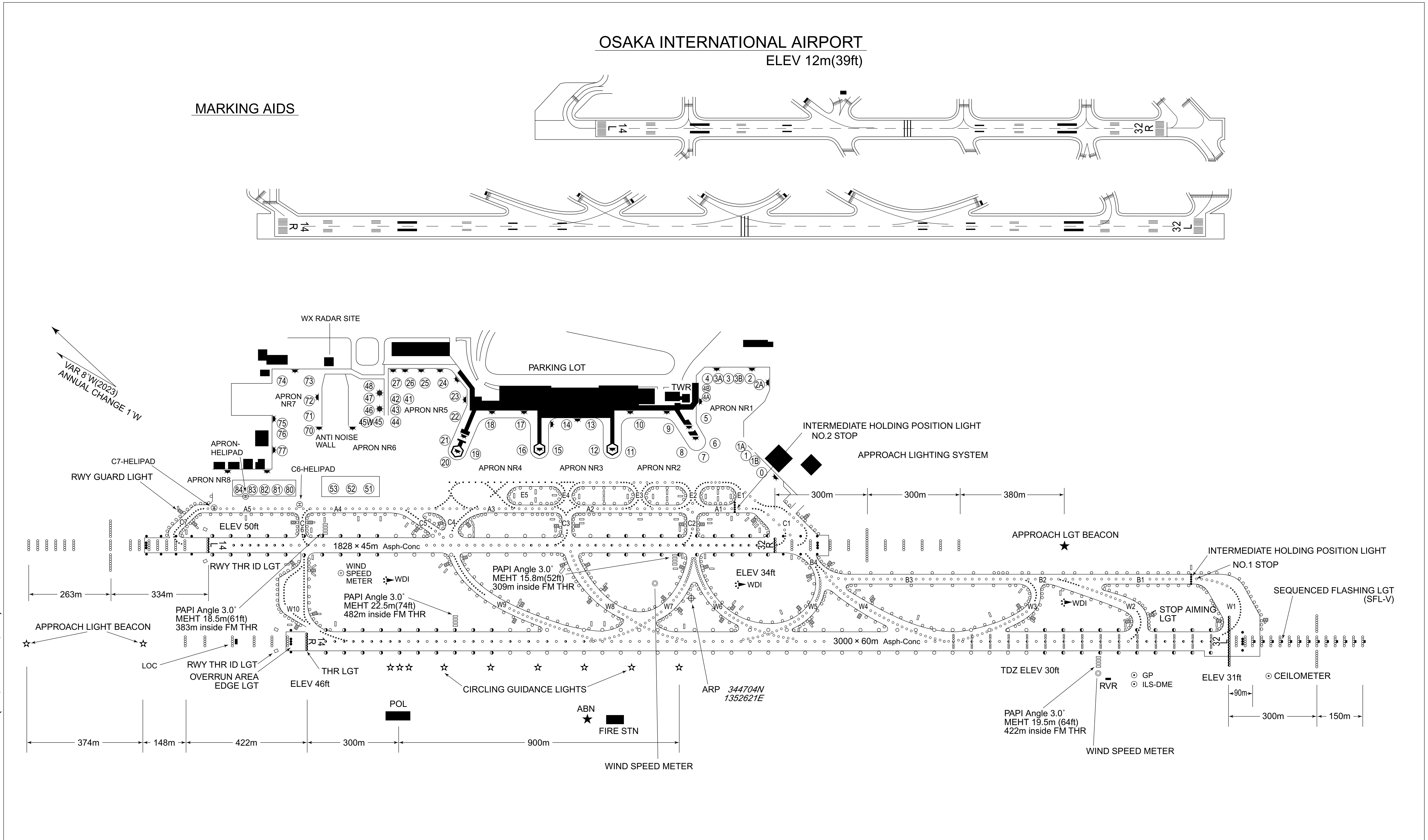
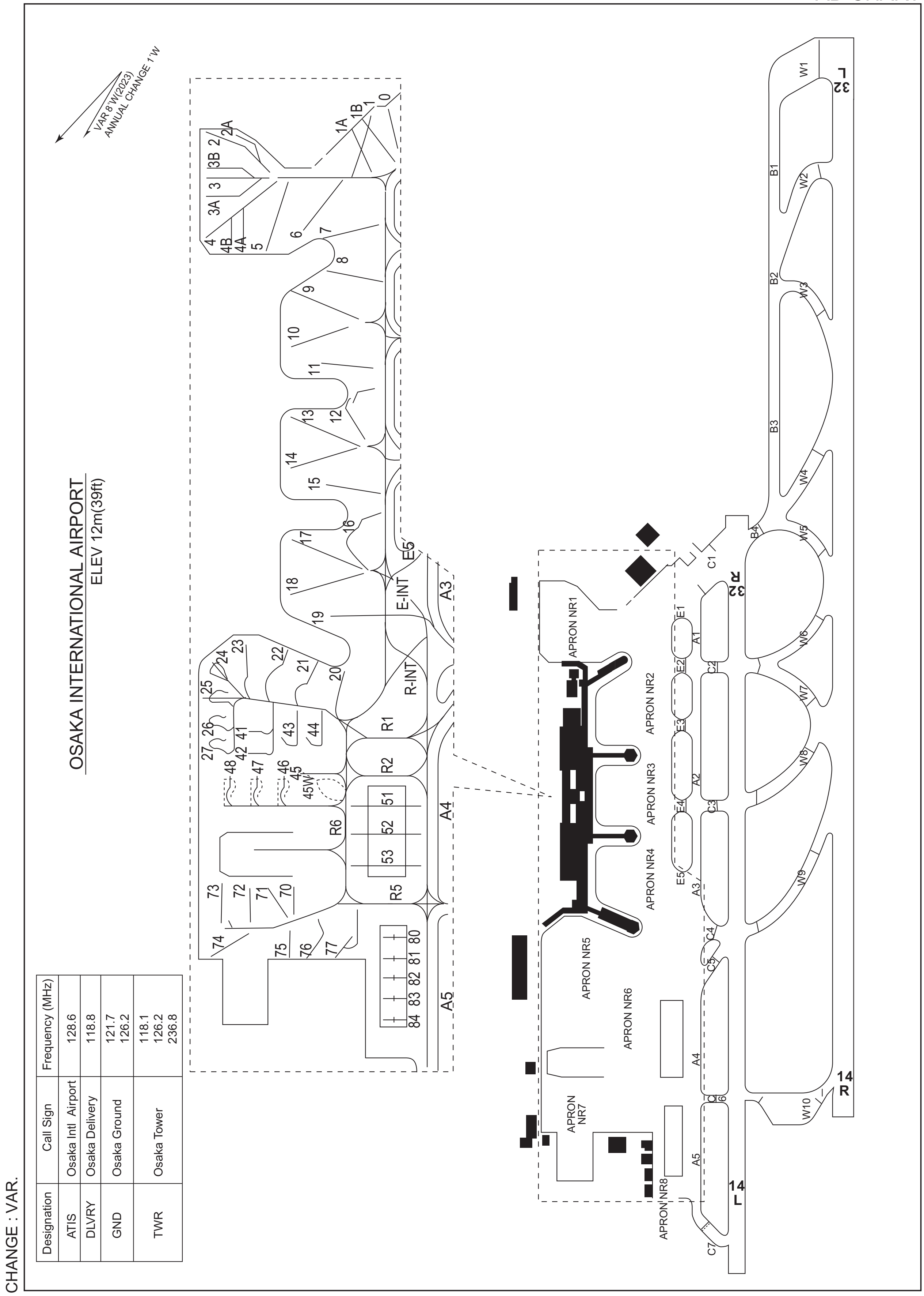
AERODROME CHART



CHANGE : HELIPAD(C6, APRON, C7) installed.

RJOO / OSAKA INTL

AD CHART



Designation	Call Sign	Frequency (MHz)
ATIS	Osaka Intl Airport	128.6
DLVRY	Osaka Delivery	118.8
GND	Osaka Ground	121.7 126.2
TWR	Osaka Tower	118.1 126.2 236.8

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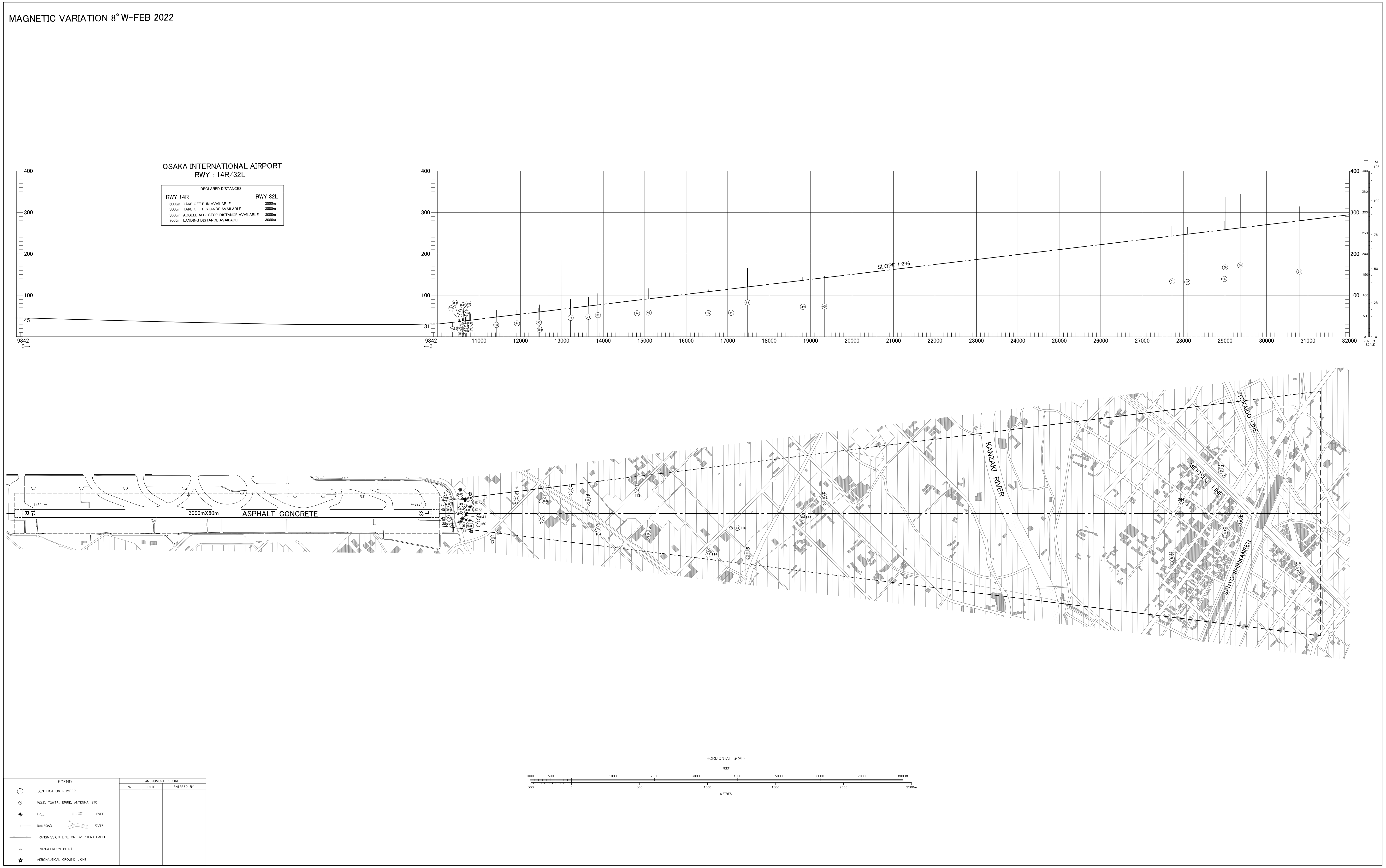
AERODROME OBSTACLE CHART-ICAO
TYPE A (OPERATING LIMITATIONS)

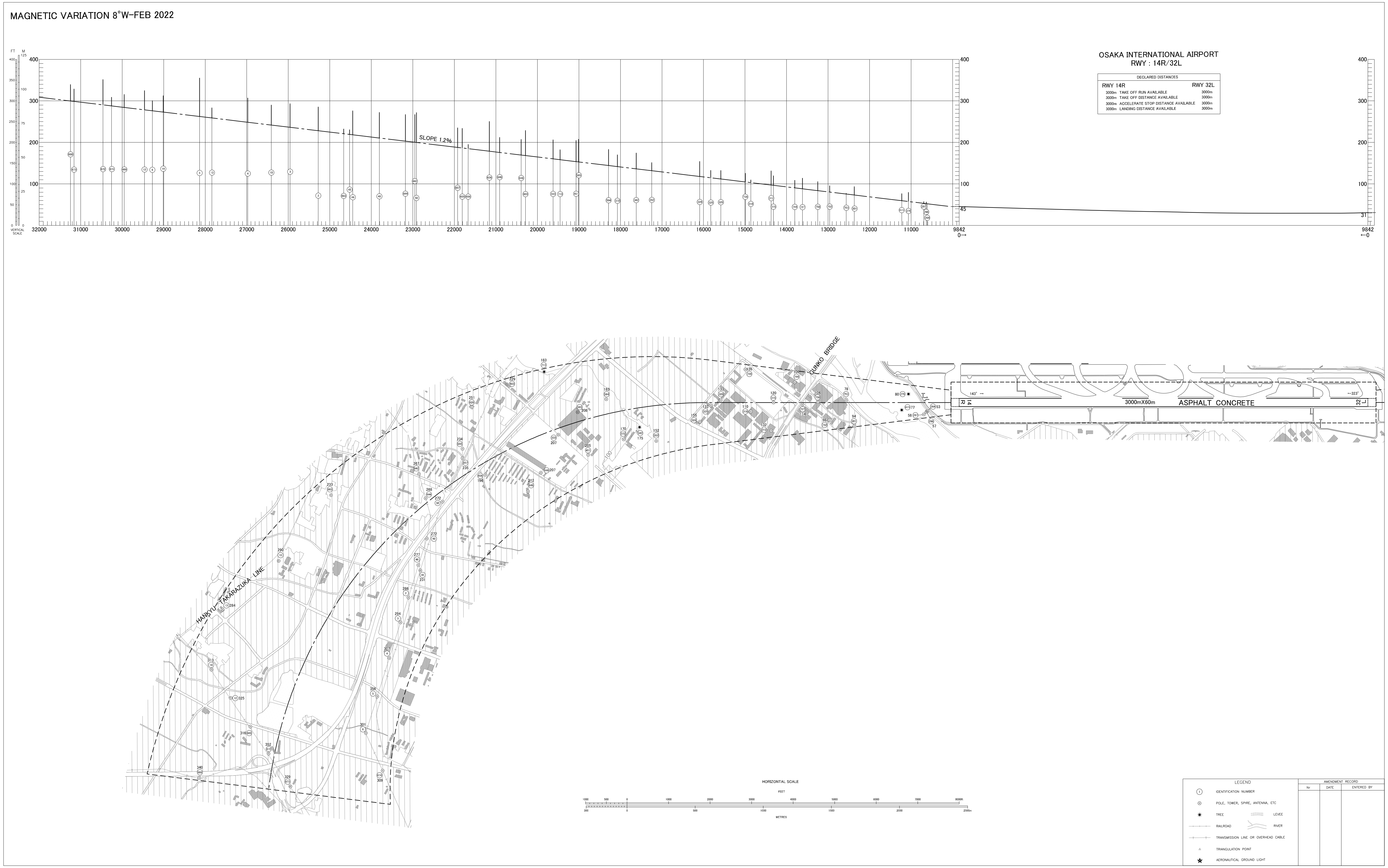
DIMENSIONS AND ELEVATIONS IN FEET BEARINGS ARE MAGNETIC



CHANGE : Update

AERODROME OBSTACLE CHART-ICAO
TYPE A (OPERATING LIMITATIONS)

DIMENSIONS AND ELEVATIONS IN FEET BEARINGS ARE MAGNETIC

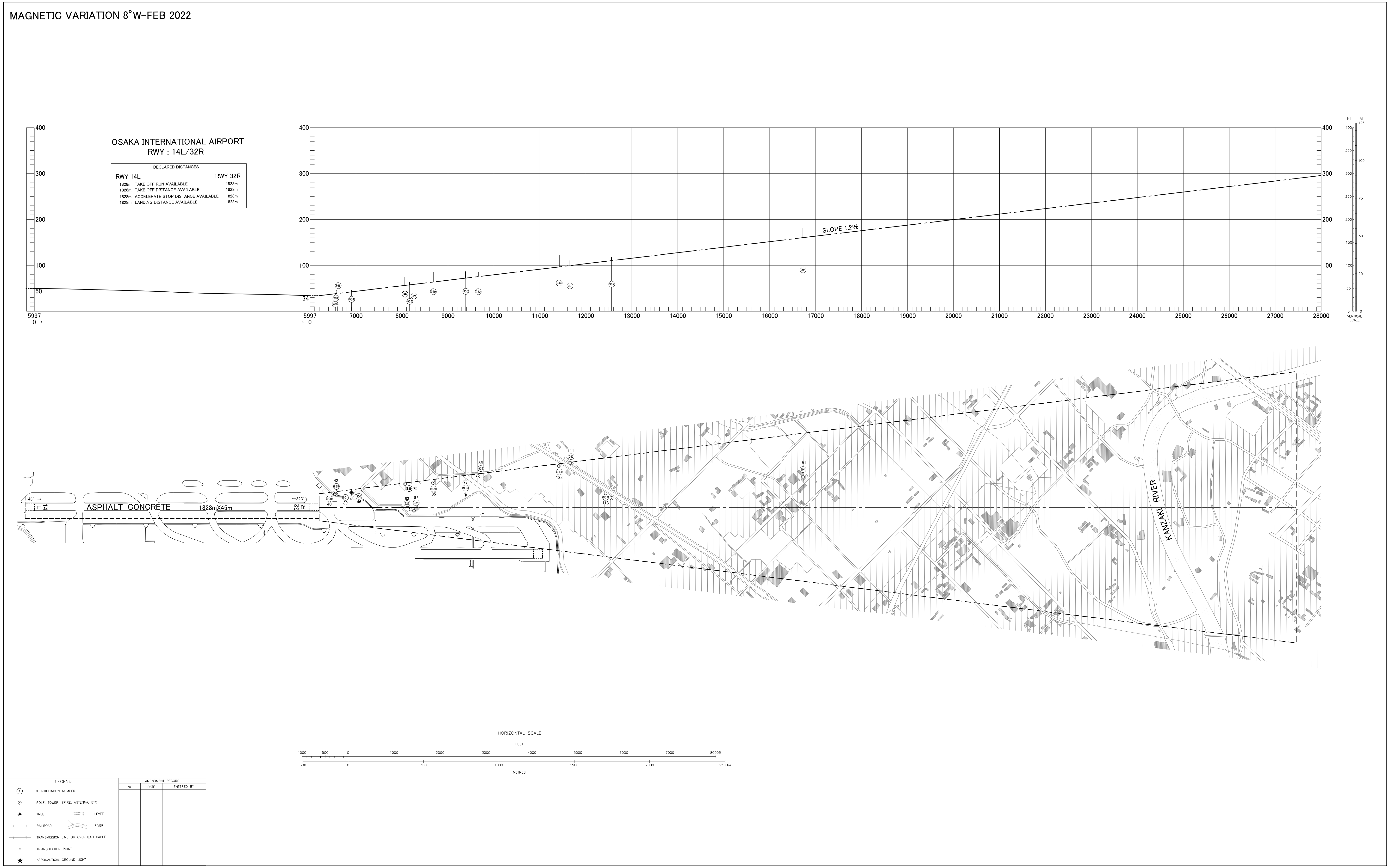


CHANGE : Update

AERODROME OBSTACLE CHART-ICAO

TYPE A (OPERATING LIMITATIONS)

DIMENSIONS AND ELEVATIONS IN FEET BEARINGS ARE MAGNETIC

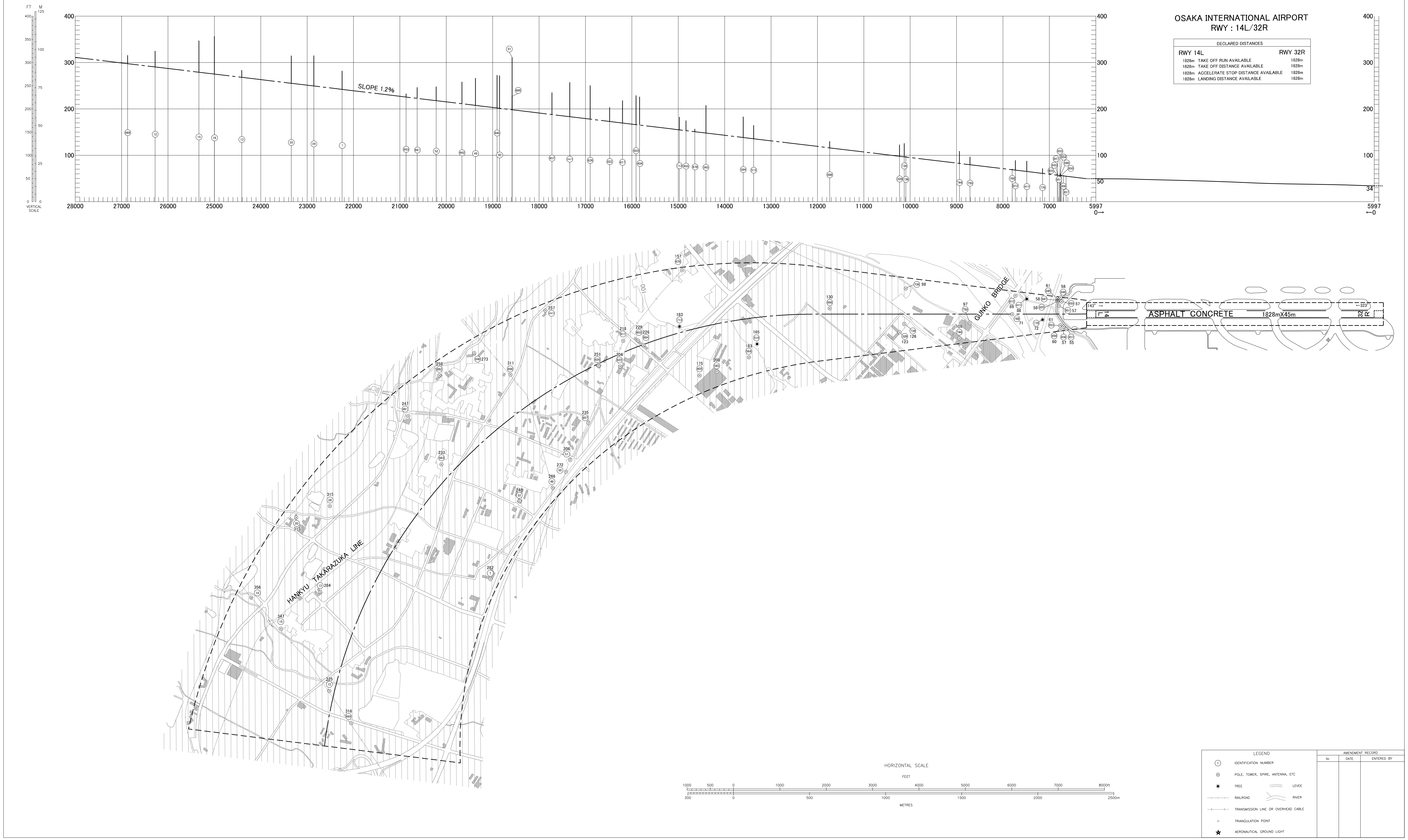


CHANGE : Update

AERODROME OBSTACLE CHART-ICAO
TYPE A (OPERATING LIMITATIONS)

DIMENSIONS AND ELEVATIONS IN FEET BEARINGS ARE MAGNETIC

MAGNETIC VARIATION 8° W-FEB 2022



CHANGE : Update

AERODROME OBSTACLE CHART-ICAO
TYPE B

DIMENSIONS AND ELEVATIONS IN FEET BEARINGS ARE MAGNETIC

CHANGE : Update

STANDARD DEPARTURE CHART -INSTRUMENT

RJOO / OSAKA INTL SID

ASUKA FOUR DEPARTURE

RWY 32R/32L : Climb RWY HDG to 500FT or above, turn left within 4NM from RWY end/ITE 2.1DME,...
RWY 14R/14L : Climb RWY HDG to 500FT or above, turn left,...
...via ITE R101 to ASUKA.
Cross ASUKA at or above 5000FT.

Note : When take off RWY 14R/14L, following climb gradient should be maintained until 500FT.

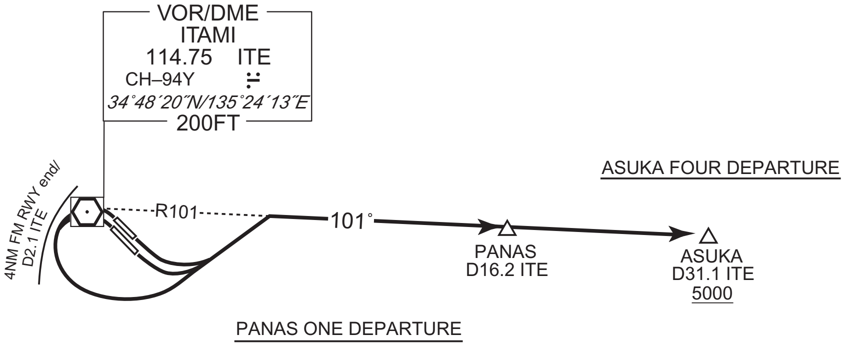
Speed (Knots)	60	90	120	150	180	210
Rate (Feet/Min)	300	450	600	750	900	1050

PANAS ONE DEPARTURE

RWY 32R/32L : Climb RWY HDG to 500FT or above, turn left within 4NM from RWY end/ITE 2.1DME,...
RWY 14R/14L : Climb RWY HDG to 500FT or above, turn left,...
...via ITE R101 to PANAS.

Note : When take off RWY 14R/14L, following climb gradient should be maintained until 500FT.

Speed (Knots)	60	90	120	150	180	210
Rate (Feet/Min)	300	450	600	750	900	1050



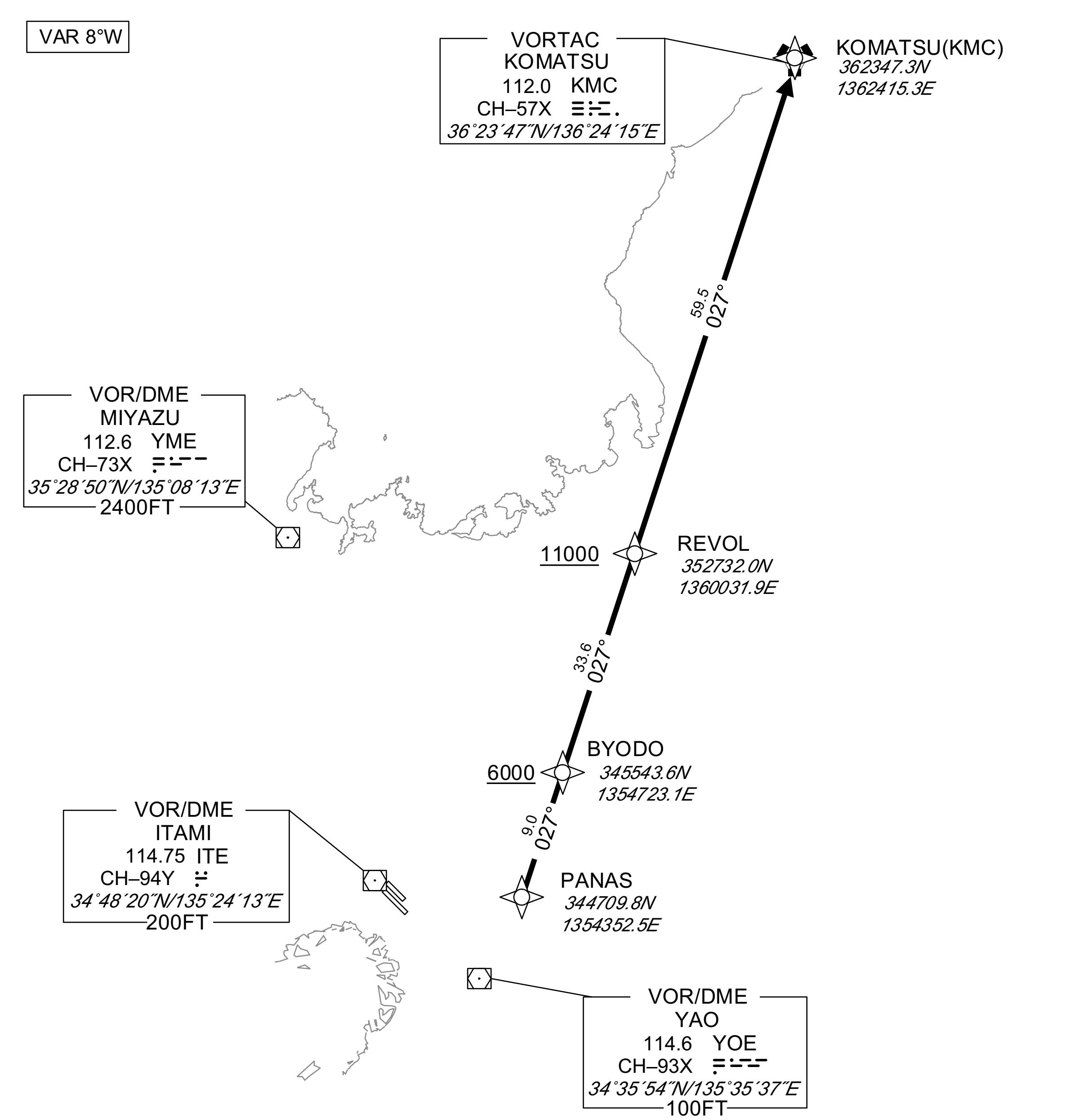
CHANGE: New PROC(PANAS ONE DEPARTURE).

STANDARD DEPARTURE CHART -INSTRUMENT

RJOO / OSAKA INTL

RNAV TRANSITION

KOMAZ TRANSITION		RNAV1
Note 1) DME/DME/IRU or GNSS required. 2) RADAR service required.	Critical DME	-
	DME GAP	-
	Inappropriate Navaids	See AD1.1.6.10.3. Inappropriate NAVAIDs for RNAV1



From PANAS, to BYODO at or above 6000FT, to REVOL at or above 11000FT, to KMC.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	PANAS	-	-	-8.0	-	-	-	-	-	RNAV1
002	TF	BYODO	-	027 (018.6)	-8.0	9.0	-	+6000	-	-	RNAV1
003	TF	REVOL	-	027 (018.6)	-8.0	33.6	-	+11000	-	-	RNAV1
004	TF	KMC	-	027 (018.7)	-8.0	59.5	-	-	-	-	RNAV1

CHANGE : Description of VAR and PROC name.

STANDARD DEPARTURE CHART -INSTRUMENT

RJOO / OSAKA INTL SID

IZUMI ONE DEPARTURE

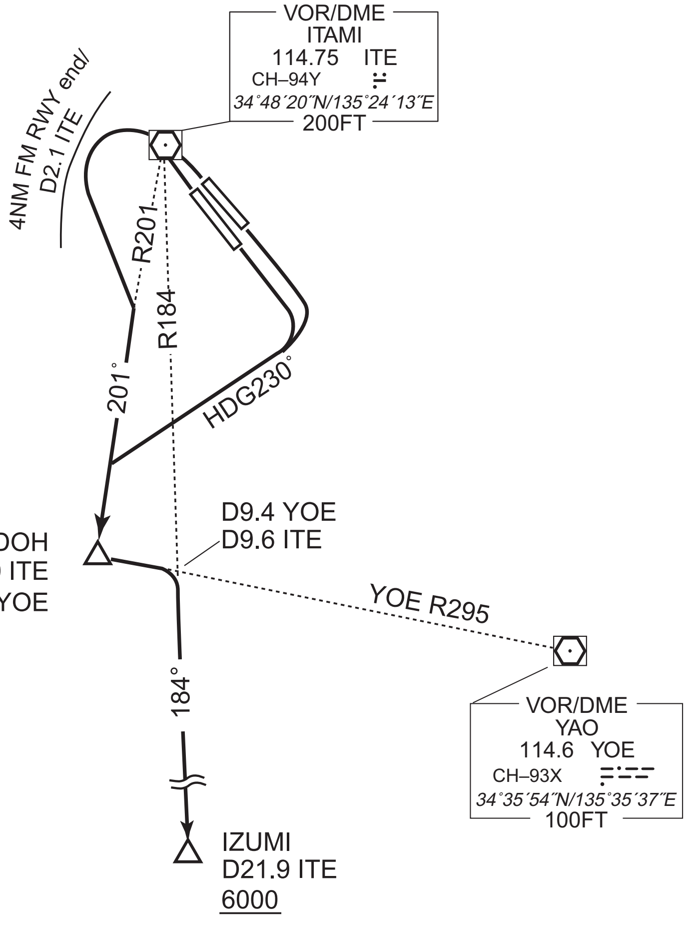
RWY 32R/32L : Climb RWY HDG to 500FT or above, turn left within 4NM from RWY end/ITE 2.1DME, via ITE R201 to YODOH,...

RWY 14R/14L : Climb RWY HDG to 500FT or above, turn right HDG230˚ to intercept and proceed via ITE R201 to YODOH,...

...turn left, via YOE R295 to intercept and proceed via ITE R184 to IZUMI. Cross IZUMI at or above 6000FT.

Note : When take off RWY 14R/14L, following climb gradient should be maintained until 500FT.

Speed (Knots)	60	90	120	150	180	210
Rate (Feet/Min)	300	450	600	750	900	1050



CHANGE : Description of PROC name.

STANDARD DEPARTURE CHART -INSTRUMENT

RJOO / OSAKA INTL SID

EAST REVERSAL FOUR DEPARTURE

RWY 32R/32L : Climb RWY HDG to 500FT or above, turn left within 4NM from RWY end/ITE 2.1DME,...

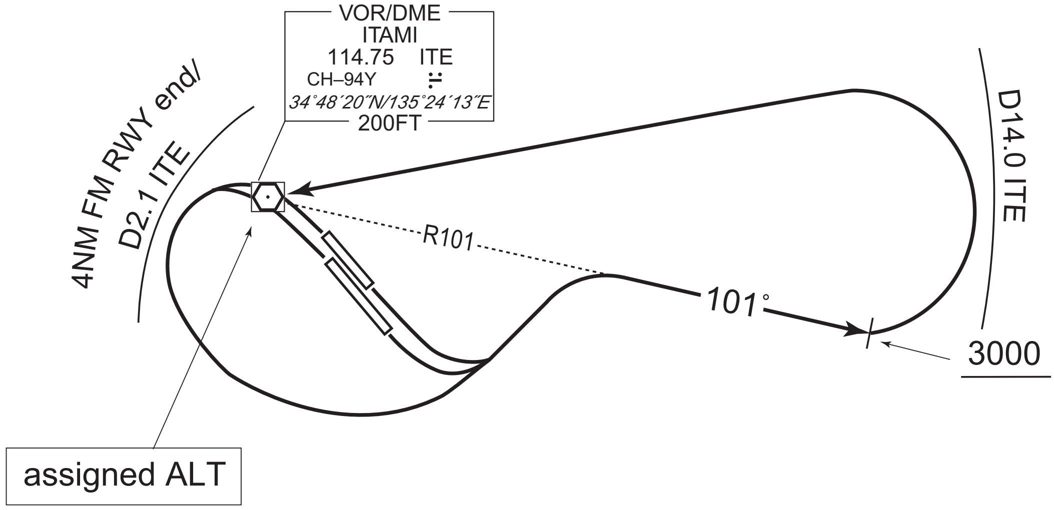
RWY 14R/14L : Climb RWY HDG to 500FT or above, turn left,...

...via ITE R101 to 3000FT or above, turn left direct to ITE VOR/DME within ITE 14.0DME.

Cross ITE VOR/DME at assigned altitude.

Note : When take off RWY 14R/14L, following climb gradient should be maintained until 500FT.

Speed (Knots)	60	90	120	150	180	210
Rate (Feet/Min)	300	450	600	750	900	1050



CHANGE : Description of PROC name.

STANDARD DEPARTURE CHART -INSTRUMENT

AIP JAPAN
RJOO / OSAKA INTL SID

TIGER TWO DEPARTURE

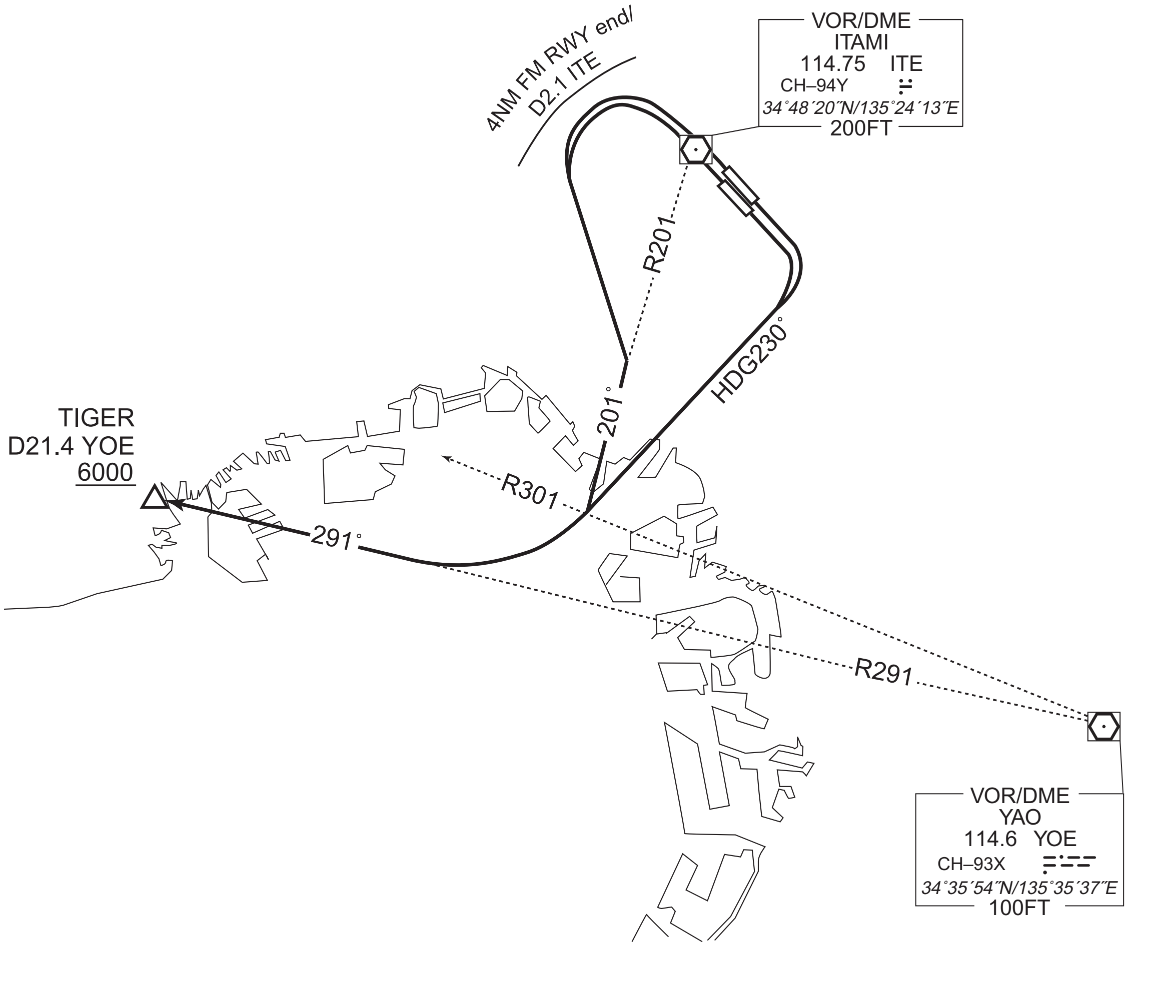
RWY 32R/32L : Climb RWY HDG to 500FT or above, turn left within 4NM from RWY end/ITE 2.1DME, via ITE R201 until crossing YOE R301...

RWY 14R/14L : Climb RWY HDG to 500FT or above, turn right HDG230˚ until crossing YOE R301...

...turn right to intercept and proceed via YOE R291 to TIGER.
Cross TIGER at or above 6000FT.

Note : Following climb gradient should be maintained until 2500FT.

Speed (Knots)	60	90	120	150	180	210
Rate (Feet/Min)	300	450	600	750	900	1050



CHANGE : Description of PROC name.

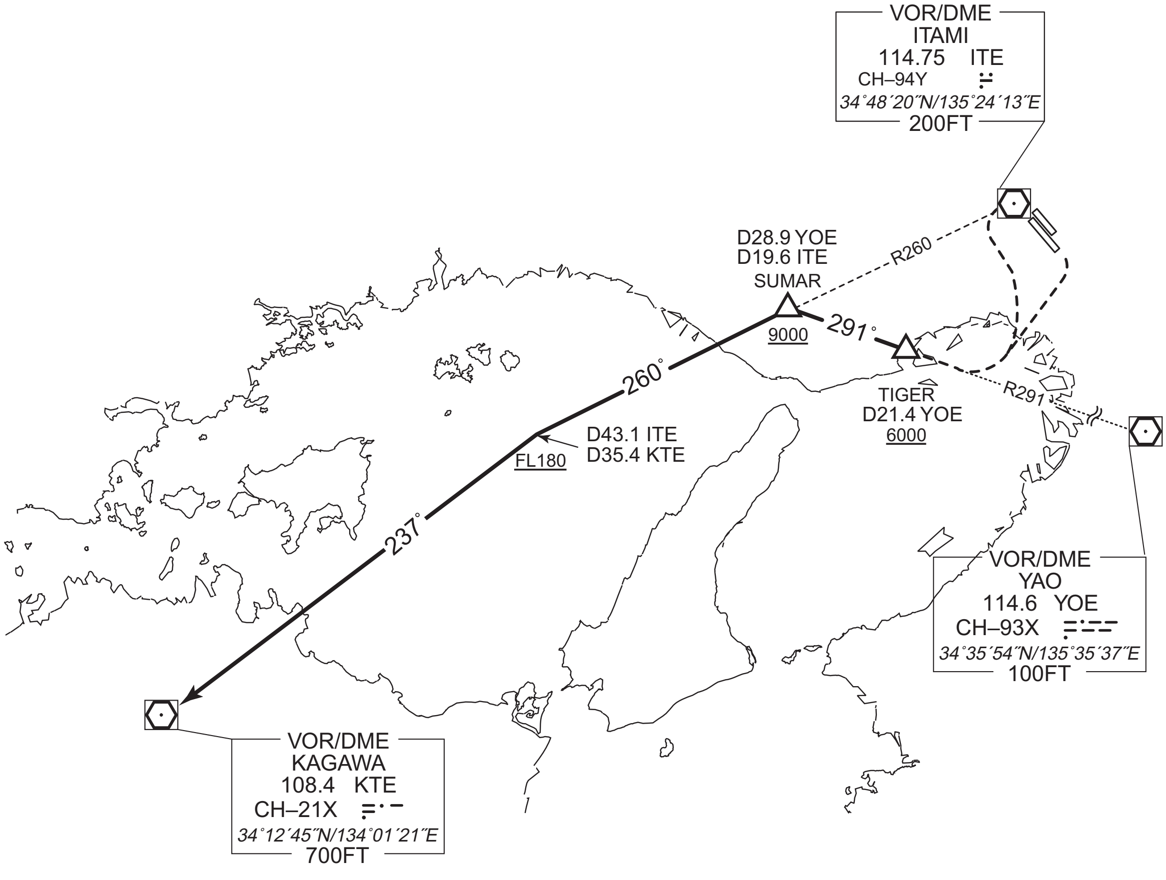
STANDARD DEPARTURE CHART -INSTRUMENT

RJOO / OSAKA INTL

TRANSITION

KAGAWA TRANSITION

From over TIGER, via YOE R291 to SUMAR, via ITE R260 to intercept and proceed via KTE R057 to KTE VOR/DME.
Cross SUMAR at or above 9000FT, cross ITE R260/43.1DME at or above FL180.



CHANGE : Description of PROC name.

STANDARD DEPARTURE CHART -INSTRUMENT

RJOO / OSAKA INTL — TRANSITION

ASAGI TRANSITION

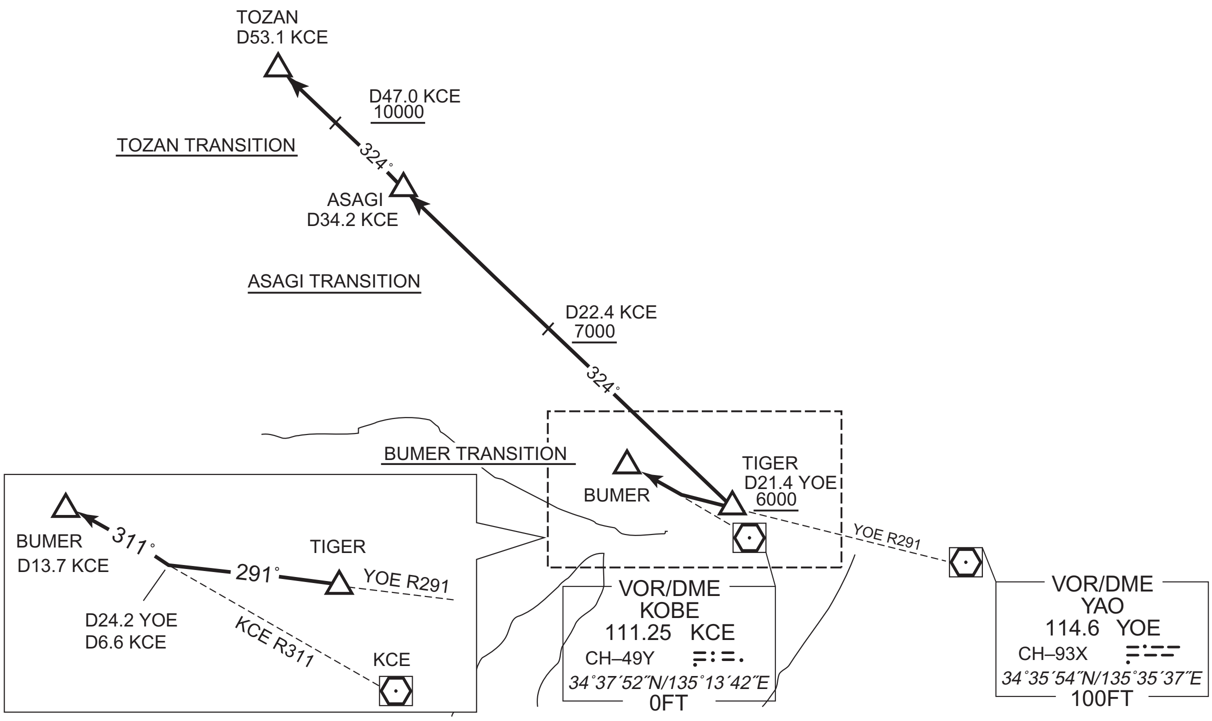
From over TIGER, via KCE R324 to ASAGI.
Cross KCE R324/22.4DME at or above 7000FT.

TOZAN TRANSITION

From over TIGER, via KCE R324 to TOZAN, via ASAGI.
Cross KCE R324/22.4DME at or above 7000FT, cross KCE R324/47.0DME at or above 10000FT.

BUMER TRANSITION

From over TIGER, via YOE R291 to intercept and proceed via KCE R311 to BUMER.



CHANGE : TOZAN TRANSITION. Radial FM KCE.

STANDARD DEPARTURE CHART-INSTRUMENT

RJOO / OSAKA INTL

SID and TRANSITION

MINAC FOUR DEPARTURE

RWY 32R/32L : Climb RWY HDG to 500FT or above, turn left within 4NM from RWY end/ITE 2.1DME,...

RWY 14R/14L : Climb RWY HDG to 500FT or above, turn left,...

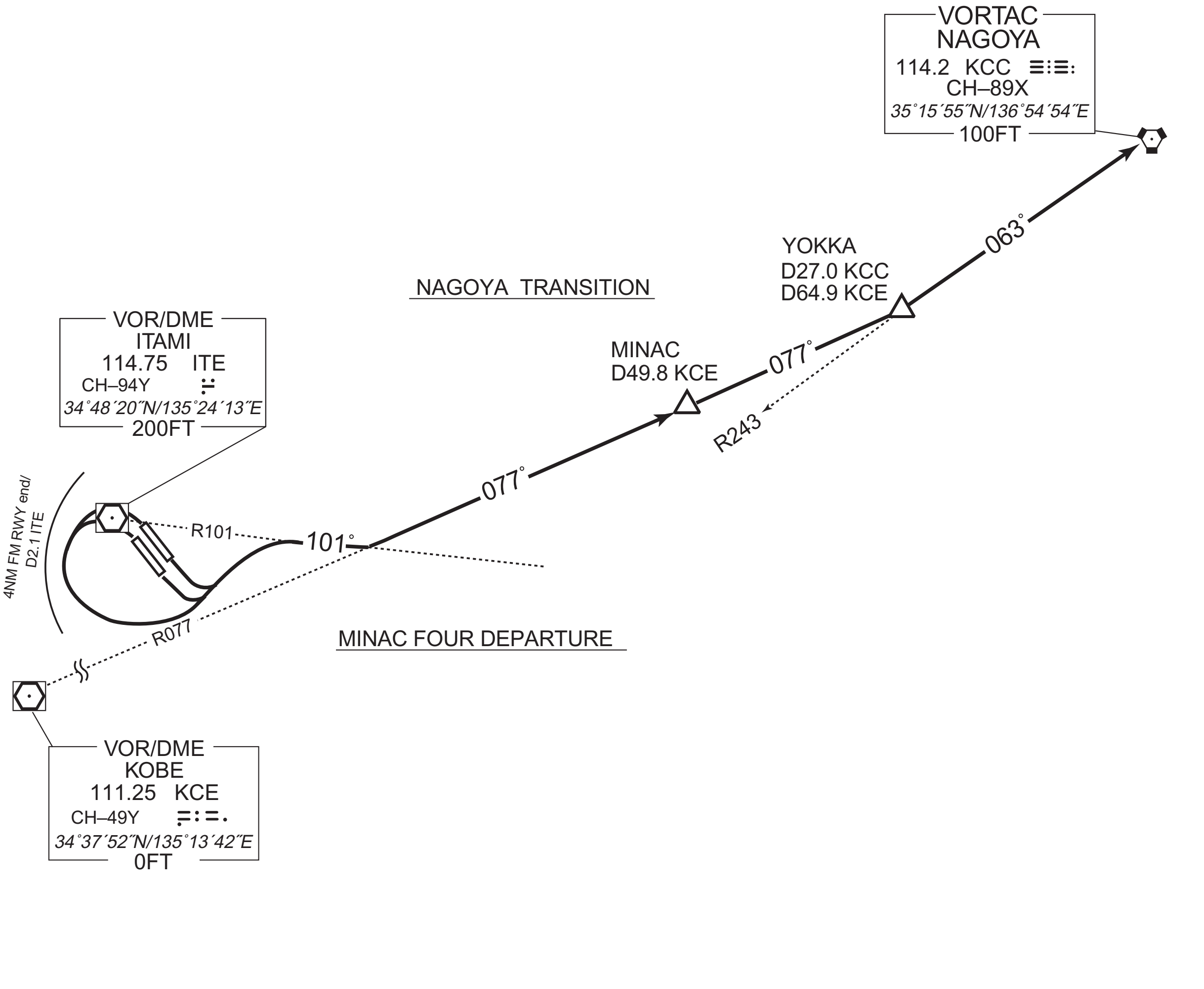
...via ITE R101 to intercept and proceed via KCE R077 to MINAC.

Note: When take off RWY14R/14L, following climb gradient should be maintained until 500FT.

Speed (Knots)	60	90	120	150	180	210
Rate (Feet/Min)	300	450	600	750	900	1050

NAGOYA TRANSITION

From over MINAC, via KCE R077 to YOKKA, via KCC R243 to KCC VORTAC.



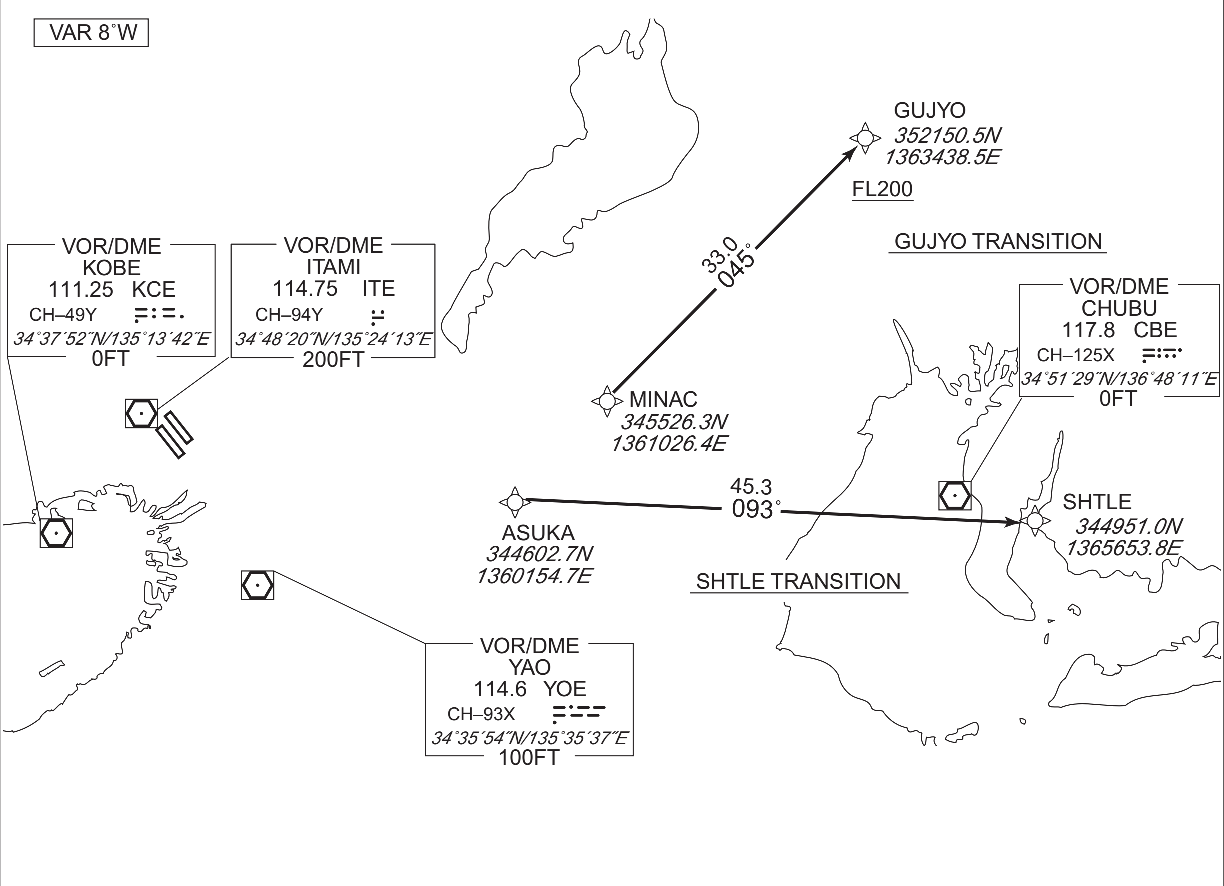
CHANGE : PROC renamed. Radial FM KCE.

STANDARD DEPARTURE CHART-INSTRUMENT

RJOO / OSAKA INTL

RNAV TRANSITION

GUJYO TRANSITION / SHTLE TRANSITION			RNAV1
NOTE 1) DME/DME/IRU or GNSS required. 2) RADAR service required.	Critical DME	–	
	DME GAP	–	
	Inappropriate Navaids	See AD1.1.6.10.3. Inappropriate NAVAIDs for RNAV1	



GUJYO TRANSITION

From MINAC, to GUJYO at or above FL200.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course ˚M(˚T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	MINAC	–	–	-8.0	–	–	–	–	–	RNAV1
002	TF	GUJYO	–	045 (036.7)	-8.0	33.0	–	+FL200	–	–	RNAV1

SHTLE TRANSITION

From ASUKA, to SHTLE.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course ˚M(˚T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	ASUKA	–	–	-8.0	–	–	–	–	–	RNAV1
002	TF	SHTLE	–	093 (084.9)	-8.0	45.3	–	–	–	–	RNAV1

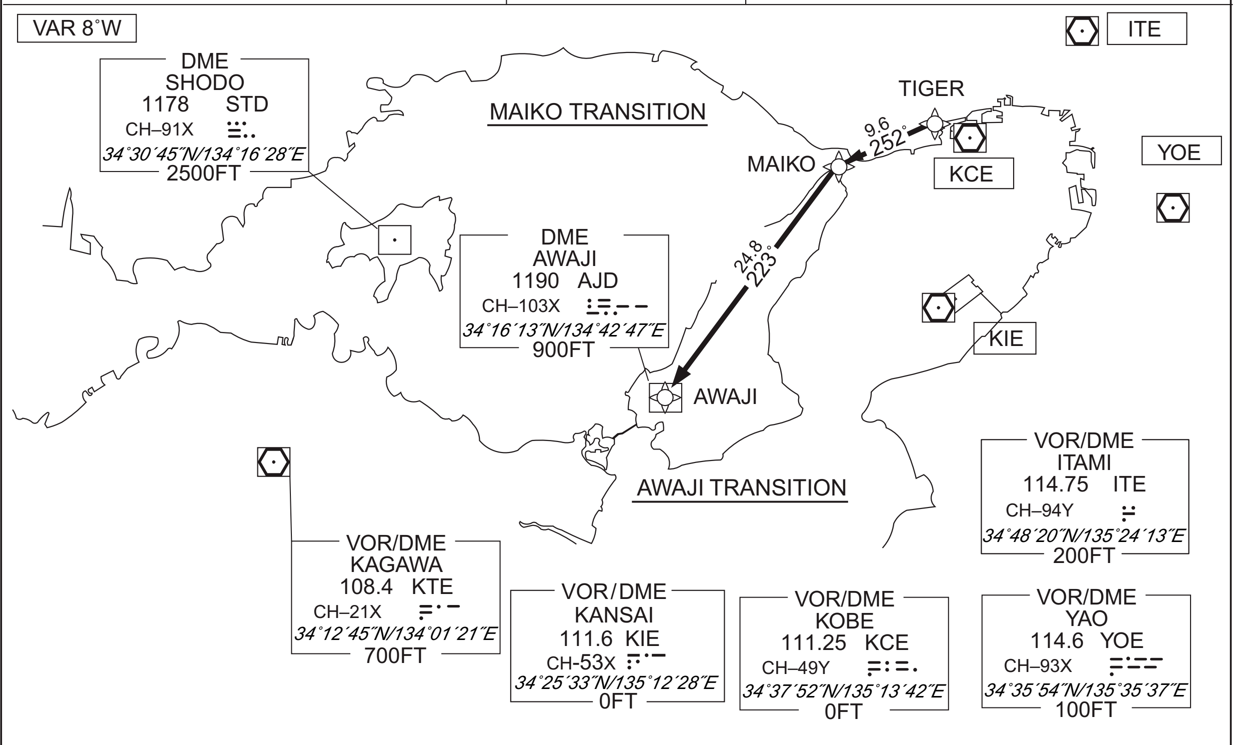
CHANGE : Description of VAR.

STANDARD DEPARTURE CHART-INSTRUMENT

RJOO / OSAKA INTL

RNAV TRANSITION

MAIKO TRANSITION / AWAJI TRANSITION		RNAV1
NOTE 1) DME/DME/IRU or GNSS required. 2) RADAR service required.	Critical DME	–
	DME GAP	–
	Inappropriate Navaids	See AD1.1.6.10.3. Inappropriate NAVAIDs for RNAV1



MAIKO TRANSITION

From TIGER, to MAIKO.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	TIGER	–	–	-8.2	–	–	–	–	–	RNAV1
002	TF	MAIKO	–	252 (244.2)	-8.2	9.6	–	–	–	–	RNAV1

AWAJI TRANSITION

From TIGER, to MAIKO, to AWAJI.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	TIGER	–	–	-8.0	–	–	–	–	–	RNAV1
002	TF	MAIKO	–	252 (244.2)	-8.0	9.6	–	–	–	–	RNAV1
003	TF	AWAJI	–	223 (214.6)	-8.0	24.8	–	–	–	–	RNAV1

Waypoint Coordinates

Waypoint Identifier	Coordinates	Waypoint Identifier	Coordinates
TIGER	*344050.4N / 1351016.7E*	AWAJI	*341613.1N / 1344246.6E*
MAIKO	*343639.7N / 1345949.1E*		

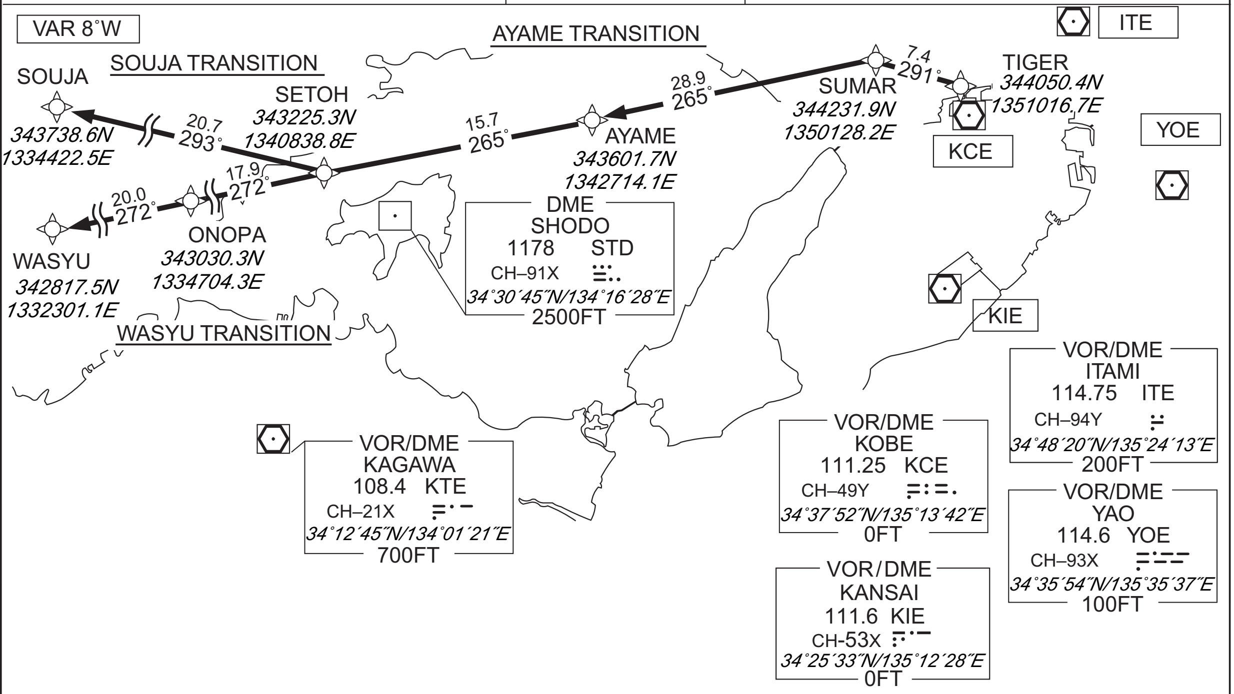
CHANGE : Longitude of STD.

STANDARD DEPARTURE CHART-INSTRUMENT

RJOO / OSAKA INTL

RNAV TRANSITION

AYAME TRANSITION / SOUJA TRANSITION / WASYU TRANSITION		RNAV1
NOTE 1) DME/DME/IRU or GNSS required. 2) RADAR service required.	Critical DME	–
	DME GAP	–
	Inappropriate Navaids	See AD1.1.6.10.3. Inappropriate NAVAIDs for RNAV1



AYAME TRANSITION

From TIGER, to SUMAR, to AYAME.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course ˚M(˚T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	TIGER	–	–	-8.1	–	–	–	–	–	RNAV1
002	TF	SUMAR	–	291 (283.2)	-8.1	7.4	–	–	–	–	RNAV1
003	TF	AYAME	–	265 (257.2)	-8.1	28.9	–	–	–	–	RNAV1

SOUJA TRANSITION

From TIGER, to SUMAR, to AYAME, to SETOH, to SOUJA.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course ˚M(˚T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	TIGER	–	–	-8.1	–	–	–	–	–	RNAV1
002	TF	SUMAR	–	291 (283.2)	-8.1	7.4	–	–	–	–	RNAV1
003	TF	AYAME	–	265 (257.2)	-8.1	28.9	–	–	–	–	RNAV1
004	TF	SETOH	–	265 (256.8)	-8.1	15.7	–	–	–	–	RNAV1
005	TF	SOUJA	–	293 (284.8)	-8.1	20.7	–	–	–	–	RNAV1

CHANGE : Longitude of STD.

STANDARD DEPARTURE CHART-INSTRUMENT

RJOO / OSAKA INTL

RNAV TRANSITION

WASYU TRANSITION

From TIGER, to SUMAR, to AYAME, to SETOH, to ONOPA, to WASYU.

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	TIGER	–	–	-8.1	–	–	–	–	–	RNAV1
002	TF	SUMAR	–	291 (283.2)	-8.1	7.4	–	–	–	–	RNAV1
003	TF	AYAME	–	265 (257.2)	-8.1	28.9	–	–	–	–	RNAV1
004	TF	SETOH	–	265 (256.8)	-8.1	15.7	–	–	–	–	RNAV1
005	TF	ONOPA	–	272 (263.9)	-8.1	17.9	–	–	–	–	RNAV1
006	TF	WASYU	–	272 (263.7)	-8.1	20.0	–	–	–	–	RNAV1

CHANGE : VAR. PROC course. ONOPA established.

STANDARD ARRIVAL CHART-INSTRUMENT

RJOO / OSAKA INTL STAR

IZUMI ARRIVAL

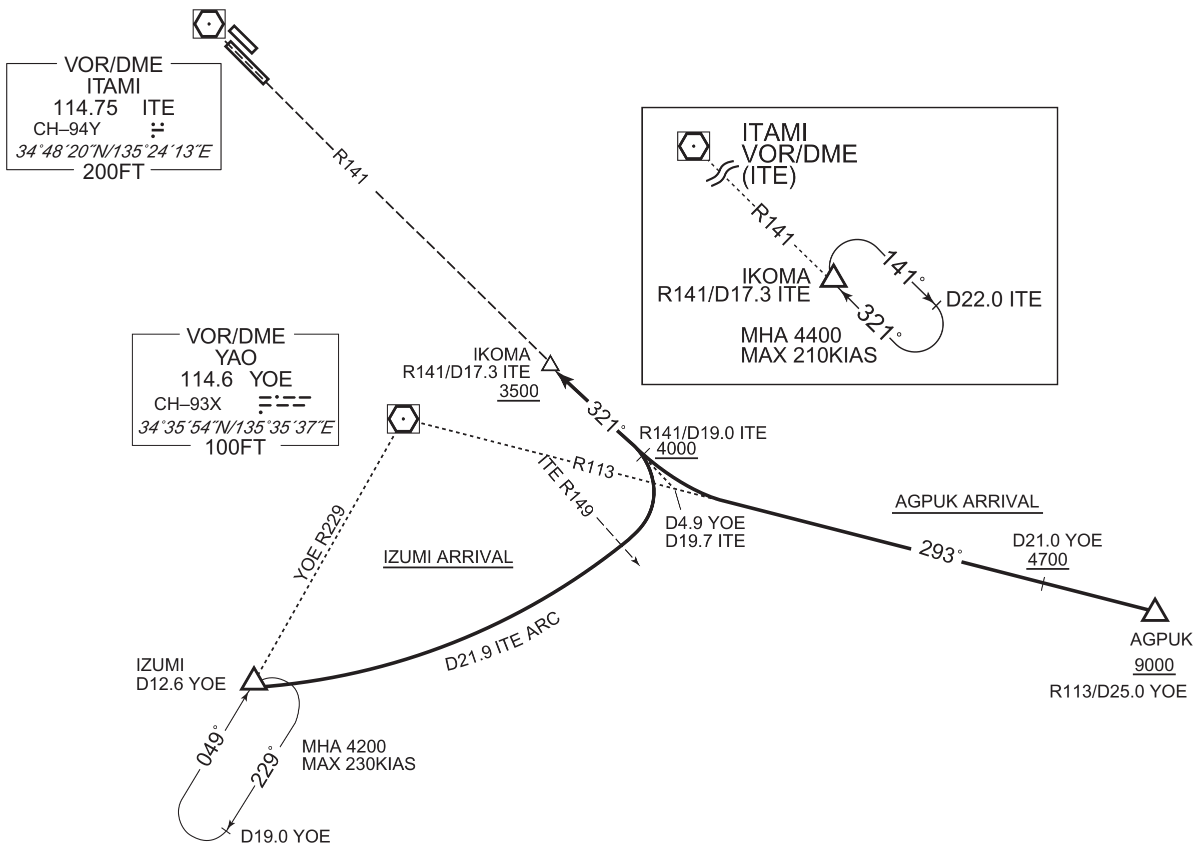
From over IZUMI, via ITE 21.9DME counterclockwise ARC to intercept and proceed via ITE R141 to IKOMA.

Cross ITE R141/19.0DME at or above 4000FT, cross IKOMA at or above 3500FT.

AGPUK ARRIVAL

From over AGPUK, via YOE R113 to intercept and proceed via ITE R141 to IKOMA.

Cross AGPUK at or above 9000FT, cross YOE R113/21.0DME at or above 4700FT, cross ITE R141/19.0DME at or above 4000FT, cross IKOMA at or above 3500FT.



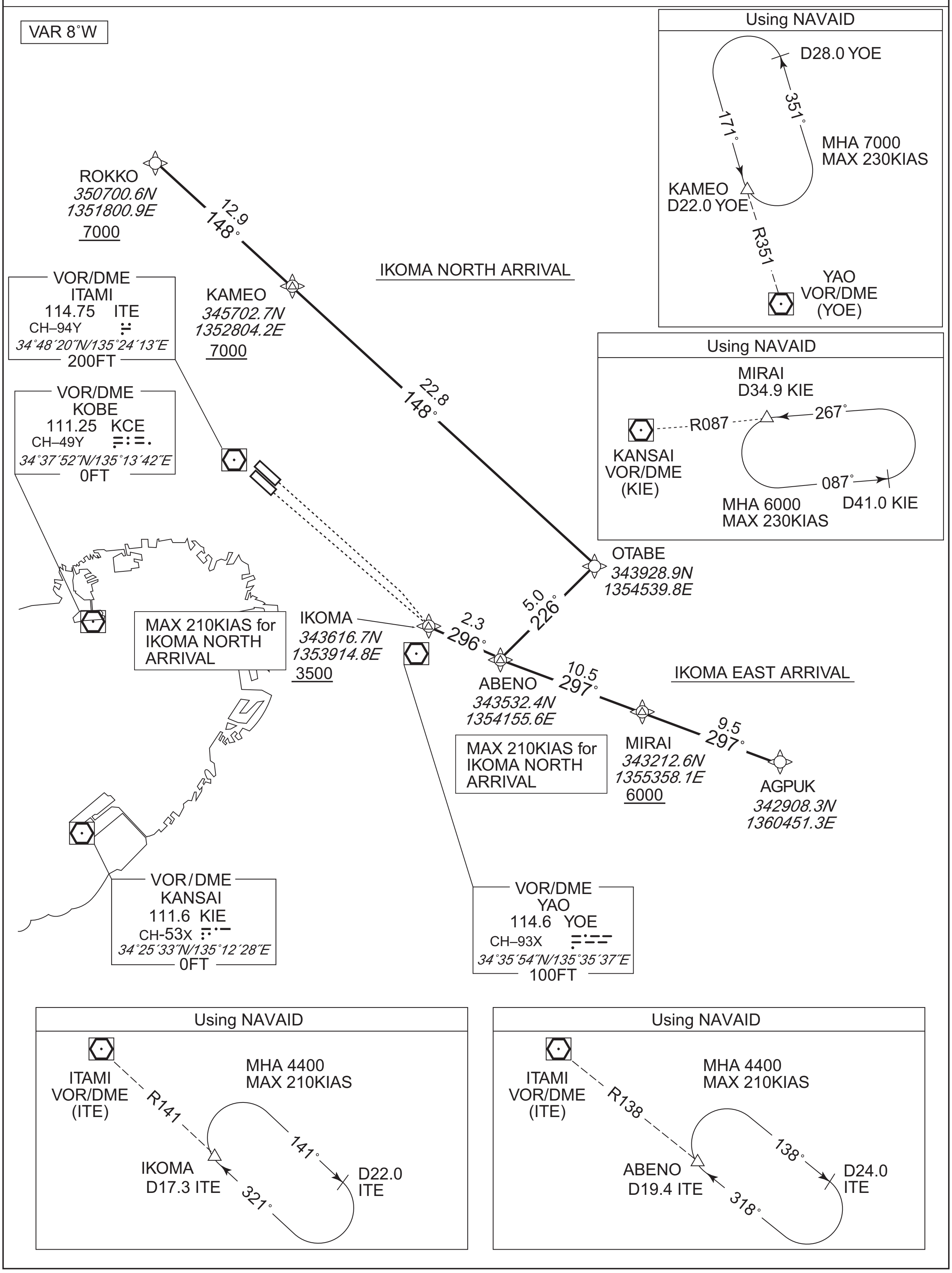
CHANGE : AGPUK ARRIVAL established.

STANDARD ARRIVAL CHART-INSTRUMENT

RJOO / OSAKA INTL — RNAV STAR RWY32L/32R

IKOMA EAST ARRIVAL / IKOMA NORTH ARRIVAL — RNAV1

Note 1) DME/DME/IRU or GNSS required
2) RADAR service required



CHANGE : Description of VAR.

STANDARD ARRIVAL CHART-INSTRUMENT

RJOO / OSAKA INTL

RNAV STAR RWY32L/32R

IKOMA EAST ARRIVAL

From AGPUK, to MIRAI at or above 6000FT, to ABENO, to IKOMA at or above 3500FT.

Critical DME	KCC : AGPUK – MIRAI
DME GAP	–
Inappropriate Navaids	See AD1.1.6.10.3. Inappropriate NAVAIDs for RNAV1

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	AGPUK	–	–	-8.0	–	–	–	–	–	RNAV1
002	TF	MIRAI	–	297 (288.7)	-8.0	9.5	–	+6000	–	–	RNAV1
003	TF	ABENO	–	297 (288.6)	-8.0	10.5	–	–	–	–	RNAV1
004	TF	IKOMA	–	296 (288.5)	-8.0	2.3	–	+3500	–	–	RNAV1

IKOMA NORTH ARRIVAL

From ROKKO at or above 7000FT, to KAMEO at or above 7000FT, to OTABE, to ABENO, to IKOMA at or above 3500FT.

Critical DME	ITE : 9.9NM to KAMEO – KAMEO YME : 19.7NM to OTABE – 13.7NM to OTABE
DME GAP	–
Inappropriate Navaids	See AD1.1.6.10.3. Inappropriate NAVAIDs for RNAV1

Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course °M(°T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	ROKKO	–	–	-8.0	–	–	+7000	–	–	RNAV1
002	TF	KAMEO	–	148 (140.4)	-8.0	12.9	–	+7000	–	–	RNAV1
003	TF	OTABE	–	148 (140.5)	-8.0	22.8	–	–	–	–	RNAV1
004	TF	ABENO	–	226 (218.0)	-8.0	5.0	–	–	-210	–	RNAV1
005	TF	IKOMA	–	296 (288.5)	-8.0	2.3	–	+3500	-210	–	RNAV1

CHANGE : VAR. KODAI abolished. AGPUK established. PROC course.

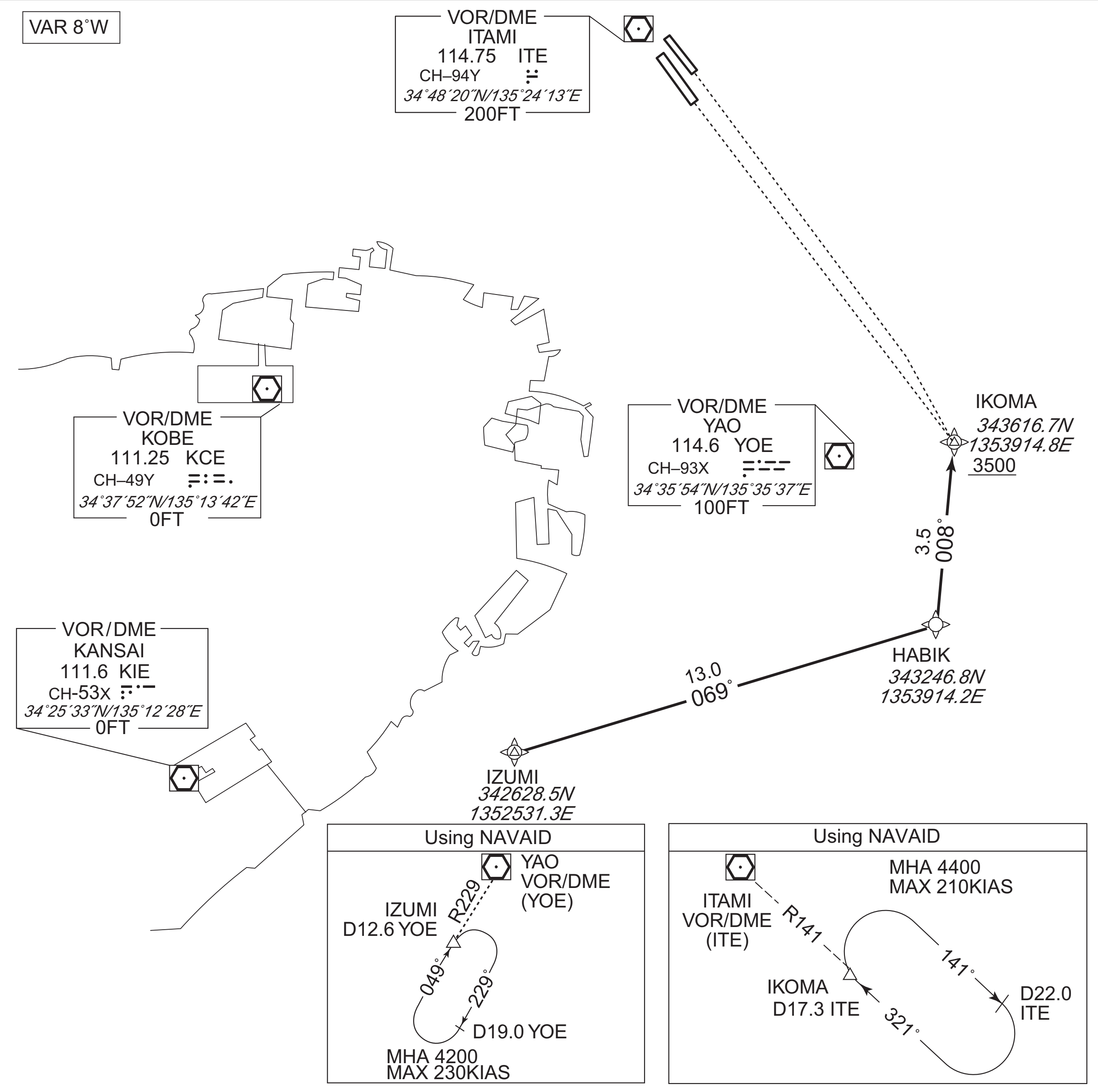
STANDARD ARRIVAL CHART-INSTRUMENT

RJOO / OSAKA INTL

RNAV STAR RWY32L/32R

HABIK ARRIVAL	RNAV1

Note 1) DME/DME/IRU or GNSS required
2) RADAR service required



From IZUMI, to HABIK, to IKOMA at or above 3500FT.

Critical DME	–
DME GAP	–
Inappropriate Navaids	See AD1.1.6.10.3. Inappropriate NAVAIDs for RNAV1

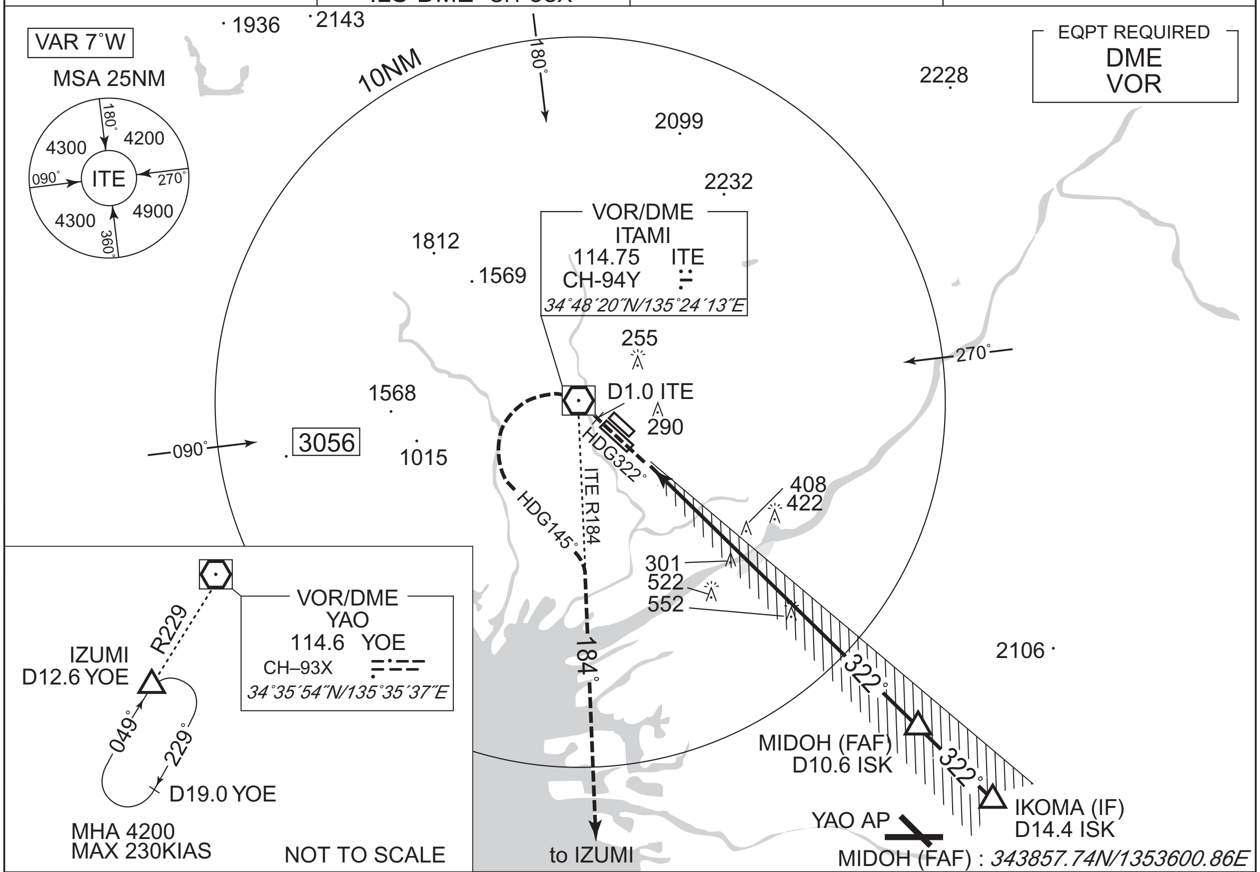
Serial Number	Path Descriptor	Waypoint Identifier	Fly Over	Course ˚M(˚T)	Magnetic Variation	Distance (NM)	Turn Direction	Altitude (FT)	Speed (KIAS)	Vertical Angle	Navigation Specification
001	IF	IZUMI	–	–	-8.0	–	–	–	–	–	RNAV1
002	TF	HABIK	–	069 (060.8)	-8.0	13.0	–	–	–	–	RNAV1
003	TF	IKOMA	–	008 (000.1)	-8.0	3.5	–	+3500	–	–	RNAV1

CHANGE : Description of VAR and PROC name.

INSTRUMENT APPROACH CHART

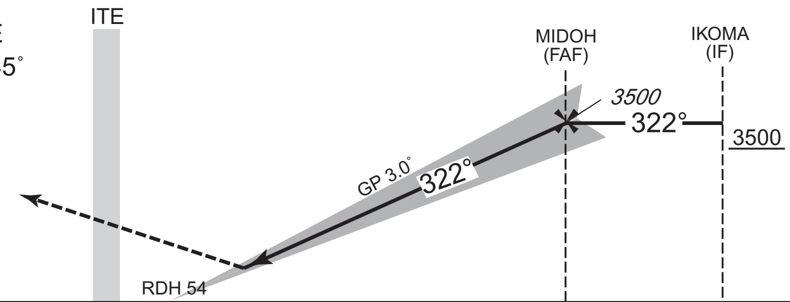
RJOO / OSAKA INTL — ILS RWY32L

KANSAI APP	ILS-LOC	OSAKA TOWER	RADAR AVBL
120.45 - 124.7 261.2	110.1 ISK ILS-GP 334.4 ILS-DME CH-38X	118.1 - 236.8 126.2 - 121.7G	ATIS 128.6



MISSED APPROACH

Climb to 5000FT on HDG322˚, 1.0DME prior to ITE VOR/DME, turn left HDG145˚ to intercept and proceed via ITE R184 to IZUMI and hold.
Contact KANSAI APP.



DME to ISK	0.2	10.6	14.4
NM to THR	0	10.4	14.2

Missed APCH climb gradient MNM 4.0%

MINIMA THR elev. 31 AD elev. 39

CAT	CAT I DA(H)	CAT I RVR/CMV	CIRCLING MDA(H) TOTAL AREA	CIRCLING MDA(H) WEST of RWY	CIRCLING VIS
A	281 (250)	700	590 (551)	590 (551)	1600
B					
C			660 (621)	610 (571)	2400
D			-	760 (721)	3200

MINIMA with Missed APCH climb gradient of 2.5% are not established.
JET circling to WEST side of RWY only.

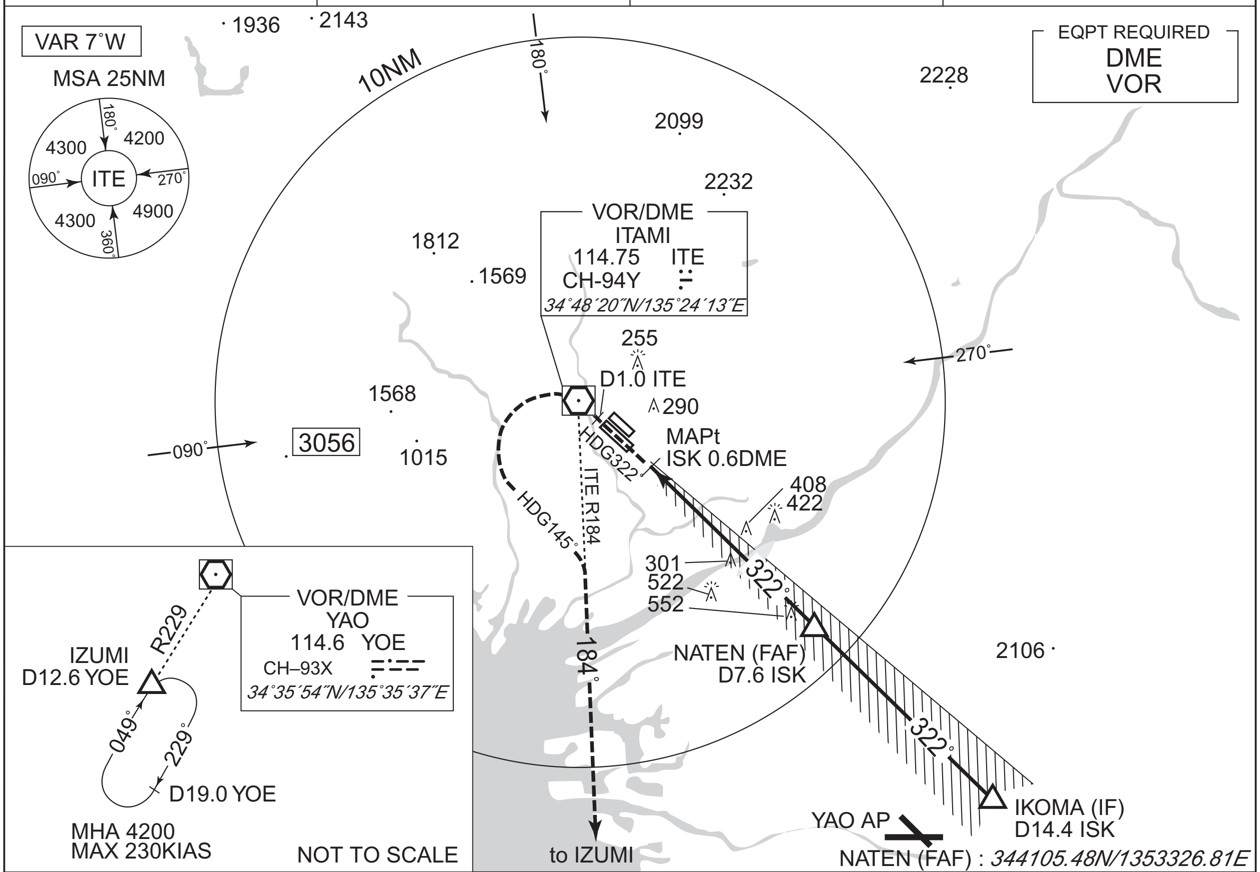
CHANGE : MDA(H) for Circling(TOTAL AREA).

INSTRUMENT APPROACH CHART

RJOO / OSAKA INTL

LOC RWY32L

KANSAI APP	ILS - LOC	OSAKA TOWER	RADAR AVBL
120.45 - 124.7 261.2	110.1 ISK ILS-DME CH-38X	118.1 - 236.8 126.2 - 121.7G	ATIS 128.6

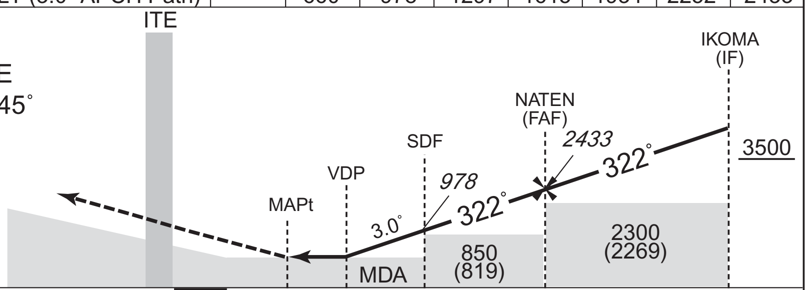


NM to ISK	MAPt	2	3	4	5	6	7	FAF
ALT (3.0° APCH Path)	–	660	978	1297	1615	1934	2252	2433

MISSED APPROACH

Climb to 5000FT on HDG322°, 1.0DME prior to ITE VOR/DME, turn left HDG145° to intercept and proceed via ITE R184 to IZUMI and hold. Contact KANSAI APP.

Timing not authorized for defining the MAPt.



DME to ISK	0.2	0.6	1.2	3.0	7.6	14.4
NM to THR	0	0.5	1.0	2.8	7.4	14.2

Missed APCH climb gradient MNM 3.2%

MINIMA THR elev. 31 AD elev. 39

CAT	MDA(H)	RVR/CMV	CIRCLING MDA(H) TOTAL AREA	CIRCLING MDA(H) WEST of RWY	CIRCLING VIS
A	390 (359)	1200	590 (551)	590 (551)	1600
B		1300			
C		1400	660 (621)	610 (571)	2400
D		1600	-	760 (721)	3200

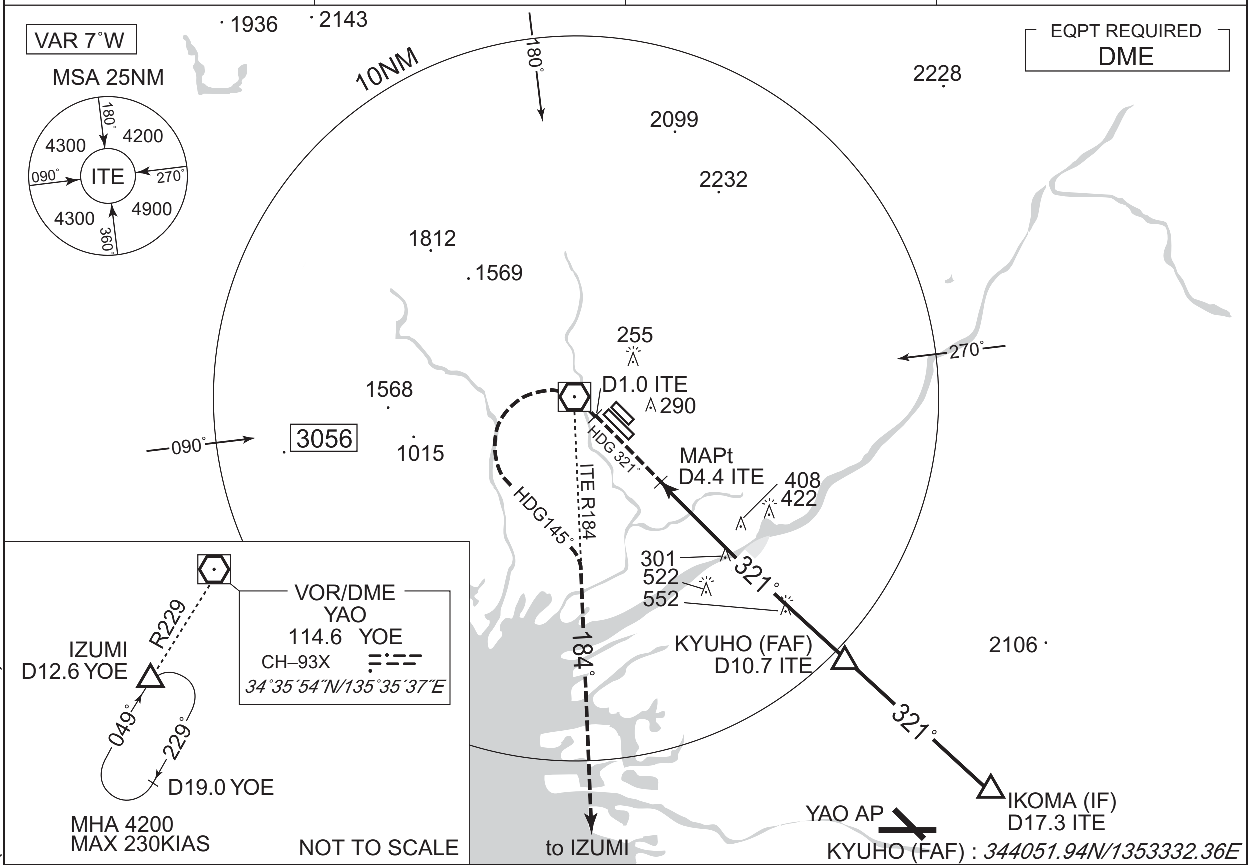
MINIMA with Missed APCH climb gradient of 2.5% are not established.
JET circling to WEST side of RWY only.

CHANGE : MDA(H) for Circling(TOTAL AREA).

INSTRUMENT APPROACH CHART

RJOO / OSAKA INTL — VOR A

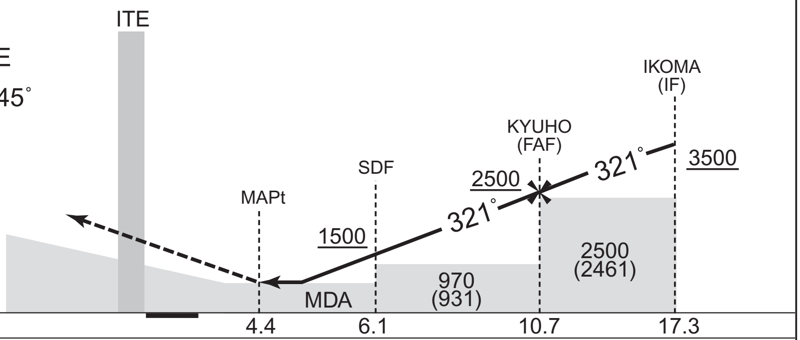
KANSAI APP	ITAMI VOR/DME	OSAKA TOWER	RADAR AVBL
120.45 - 124.7 261.2	114.75 ITE CH-94Y 34˚48´20˝N/135˚24´13˝E	118.1 - 236.8 126.2 - 121.7G	ATIS 128.6



MISSED APPROACH

Climb to 5000FT on HDG321˚, 1.0DME prior to ITE VOR/DME, turn left HDG145˚ to intercept and proceed via ITE R184 to IZUMI and hold.
Contact KANSAI APP.

Timing not authorized for defining the MAPt.



MINIMA

AD elev. 39

CAT	CIRCLING MDA(H) TOTAL AREA	CIRCLING MDA(H) WEST of RWY	VIS
A	590 (551)	590 (551)	1600
B	590 (551)	590 (551)	1600
C	660 (621)	610 (571)	2400
D	-	760 (721)	3200

JET circling to WEST side of RWY only.

CHANGE : OCA/H BTN KYUHO and SDF. MDA(H) for Circling(TOTAL AREA).

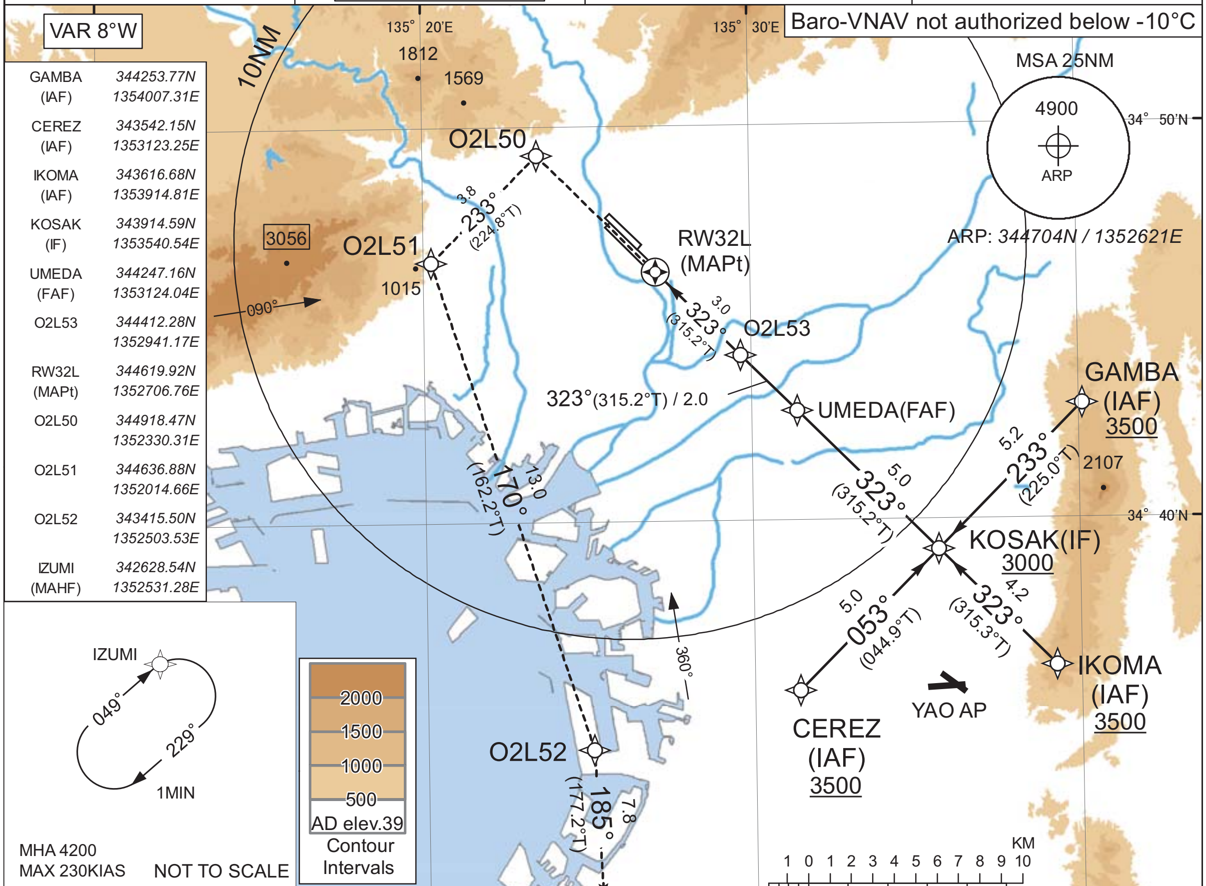
INSTRUMENT APPROACH CHART

CHANGE : O2L53 established. MSAS CH added. Minimum TEMP for Baro-VNAV. HLDG Pattern. PROC ALT at FAF. Missed APCH for Using VOR/DME abolished.
OCA/H BTN KOSAK and UMEDA. LPV established. MINIMA. NM to THR at VDP.

RJOO / OSAKA INTL — RNP RWY32L

KANSAI APP	RNP APCH	OSAKA TOWER	RADAR AVBL
120.45 - 124.7 261.2	MSAS CH68881 M32A	118.1 - 236.8 126.2 - 121.7G	ATIS 128.6

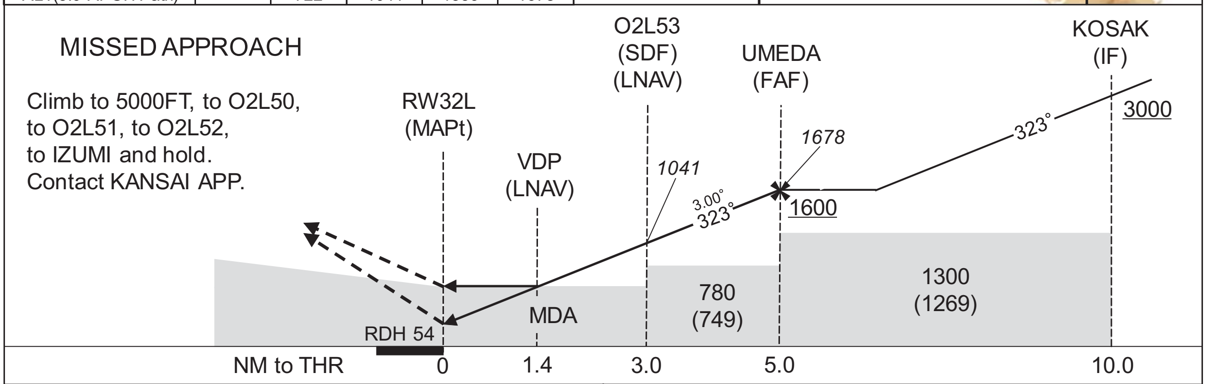
Fix	Coordinates
GAMBA (IAF)	344253.77N 1354007.31E
CEREZ (IAF)	343542.15N 1353123.25E
IKOMA (IAF)	343616.68N 1353914.81E
KOSAK (IF)	343914.59N 1353540.54E
UMEDA (FAF)	344247.16N 1353124.04E
O2L53	344412.28N 1352941.17E
RW32L (MAPt)	344619.92N 1352706.76E
O2L50	344918.47N 1352330.31E
O2L51	344636.88N 1352014.66E
O2L52	343415.50N 1352503.53E
IZUMI (MAHF)	342628.54N 1352531.28E



NM to Next Fix	MAPt	2	3	4	FAF
ALT(3.0°APCH Path)	-	722	1041	1359	1678

MISSED APPROACH

Climb to 5000FT, to O2L50, to O2L51, to O2L52, to IZUMI and hold.
Contact KANSAI APP.



Missed APCH climb gradient MNM 6.0%

MINIMA THR elev. 31 AD elev. 39

CAT	LPV DA(H)	LPV RVR/CMV	LNAV/VNAV DA(H)	LNAV/VNAV RVR/CMV	LNAV MDA(H)	LNAV RVR/CMV	CIRCLING MDA(H) TOTAL AREA	CIRCLING MDA(H) WEST of RWY	CIRCLING VIS
A	289(258)	1000	510(479)	1400	510(479)	1400	590(551)	590(551)	1600
B	299(268)	1100		1500		1500			
C	309(278)	1200		1600		1600	660(621)	610(571)	2400
D	318(287)	1400		1800		1800	-	760(721)	3200

JET circling to WEST side of RWY only.
Missed APCH climb gradient of 6.0% up to 2400FT.

INSTRUMENT APPROACH CHART

RJOO / OSAKA INTL

RNP RWY32L

FAS DATA BLOCK

Operation type	0	LTP/FTP ellipsoidal height	+00469
SBAS service provider identifier	2	FPAP latitude	*344728.7535N*
Airport identifier	RJOO	FPAP longitude	*1352543.3300E*
Runway	323	Threshold crossing height	00016.5
Approach performance designator	0	TCH units selector	1
Route indicator		Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M32A	⊿ length offset	0000
LTP/FTP latitude	*344619.8985N*	HAL	40.0
LTP/FTP longitude	*1352706.7455E*	VAL	50.0
CRC remainder	D799CA35		

Required additional data

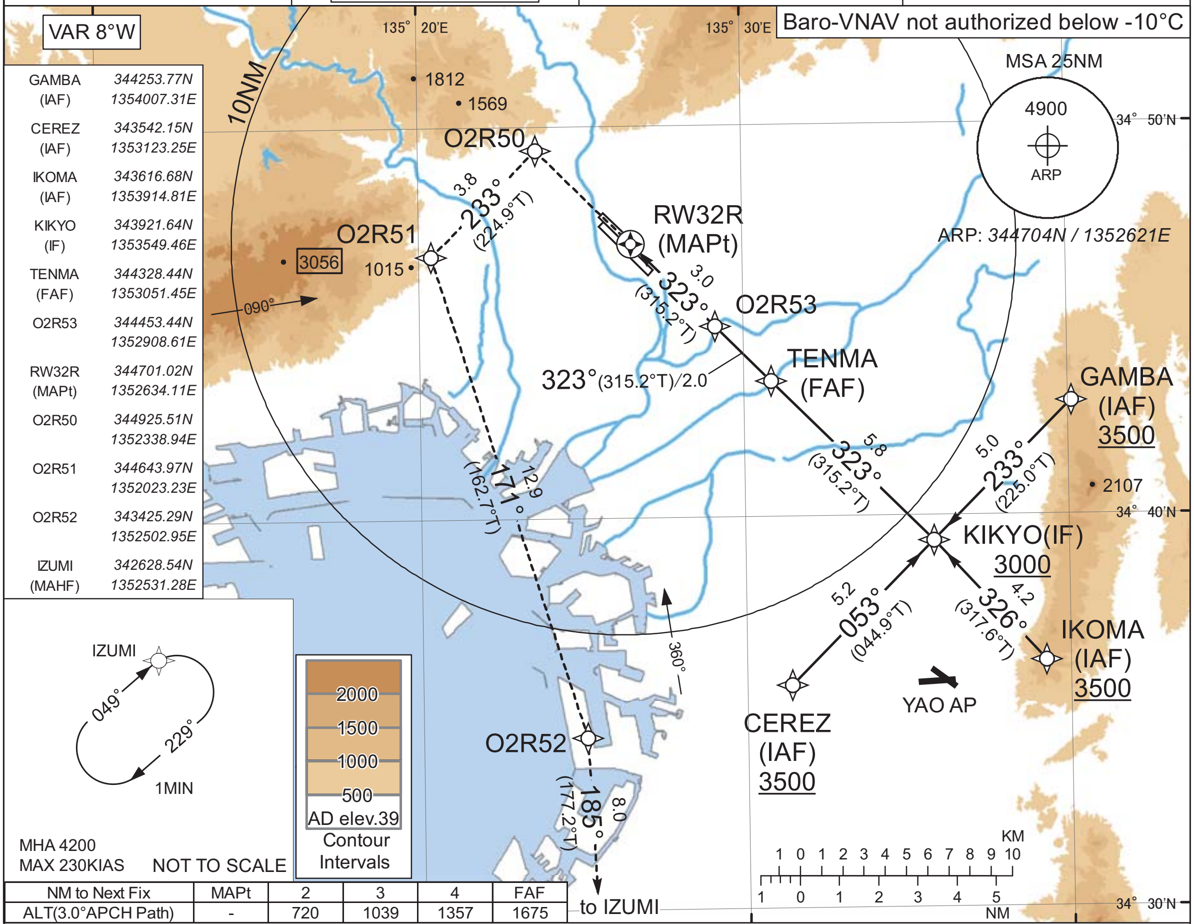
LTP/FTP orthometric height	9.6

CHANGE : FAS DATA BLOCK, Required additional data established.

INSTRUMENT APPROACH CHART

RJOO / OSAKA INTL — RNP RWY32R

KANSAI APP	RNP APCH	OSAKA TOWER	RADAR AVBL
120.45 - 124.7 261.2	MSAS CH94098 M32B	118.1 - 236.8 126.2 - 121.7G	ATIS 128.6

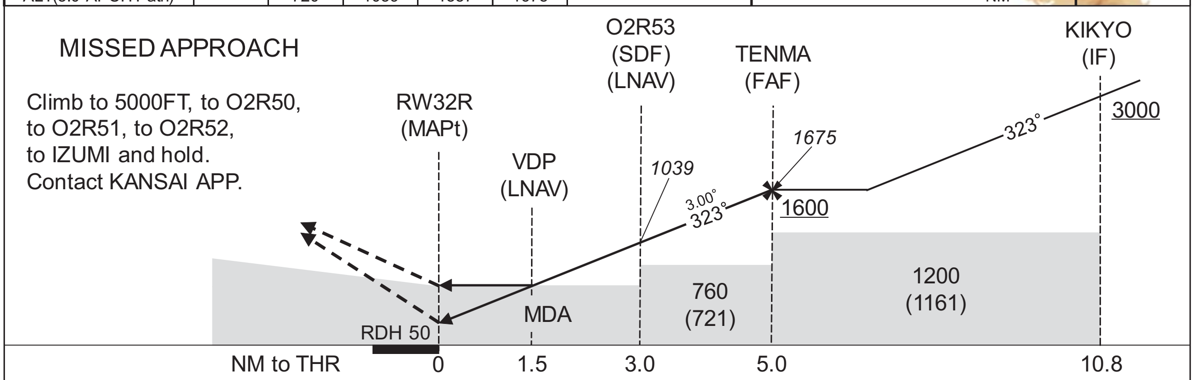


Fix	Coordinates
GAMBA (IAF)	344253.77N 1354007.31E
CEREZ (IAF)	343542.15N 1353123.25E
IKOMA (IAF)	343616.68N 1353914.81E
KIKYO (IF)	343921.64N 1353549.46E
TENMA (FAF)	344328.44N 1353051.45E
O2R53	344453.44N 1352908.61E
RW32R (MAPt)	344701.02N 1352634.11E
O2R50	344925.51N 1352338.94E
O2R51	344643.97N 1352023.23E
O2R52	343425.29N 1352502.95E
IZUMI (MAHF)	342628.54N 1352531.28E

NM to Next Fix	MAPt	2	3	4	FAF
ALT(3.0°APCH Path)	-	720	1039	1357	1675

MISSED APPROACH

Climb to 5000FT, to O2R50, to O2R51, to O2R52, to IZUMI and hold. Contact KANSAI APP.



Missed APCH climb gradient MNM 6.0%

MINIMA THR elev. 34 AD elev. 39

CAT	LPV DA(H)	LPV CMV	LNAV/VNAV DA(H)	LNAV/VNAV CMV	LNAV MDA(H)	LNAV CMV	CIRCLING MDA(H)	CIRCLING VIS
A	400(366)	1200	530(496)	1400	530(491)	1400	570(531)	1600
B	410(376)	1300		1500		1500		
C	419(385)	1400		1600		1600		2400
D	429(395)	1600		1800		1800	760(721)	3200

Circling to WEST side of RWY only.
Missed APCH climb gradient of 6.0% up to 2500FT.

CHANGE : O2R53 established. MSAS CH added. Minimum TEMP for Baro-VNAV. HLDG Pattern. ALT(3.0°APCH Path). PROC ALT.
Missed APCH for Using VOR/DME abolished. OCA/H. LPV established. MINIMA. NM to THR at VDP.

INSTRUMENT APPROACH CHART

RJOO / OSAKA INTL
RNP RWY32R

FAS DATA BLOCK

Operation type	0	LTP/FTP ellipsoidal height	+00478
SBAS service provider identifier	2	FPAP latitude	*344742.9515N*
Airport identifier	RJOO	FPAP longitude	*1352543.2590E*
Runway	321	Threshold crossing height	00015.0
Approach performance designator	0	TCH units selector	1
Route indicator		Glide path angle	03.00
Reference path data selector	0	Course width at threshold	105.00
Reference path ID	M32B	⊿ length offset	0000
LTP/FTP latitude	*344700.9970N*	HAL	40.0
LTP/FTP longitude	*1352634.0960E*	VAL	50.0
CRC remainder	49D4256C		

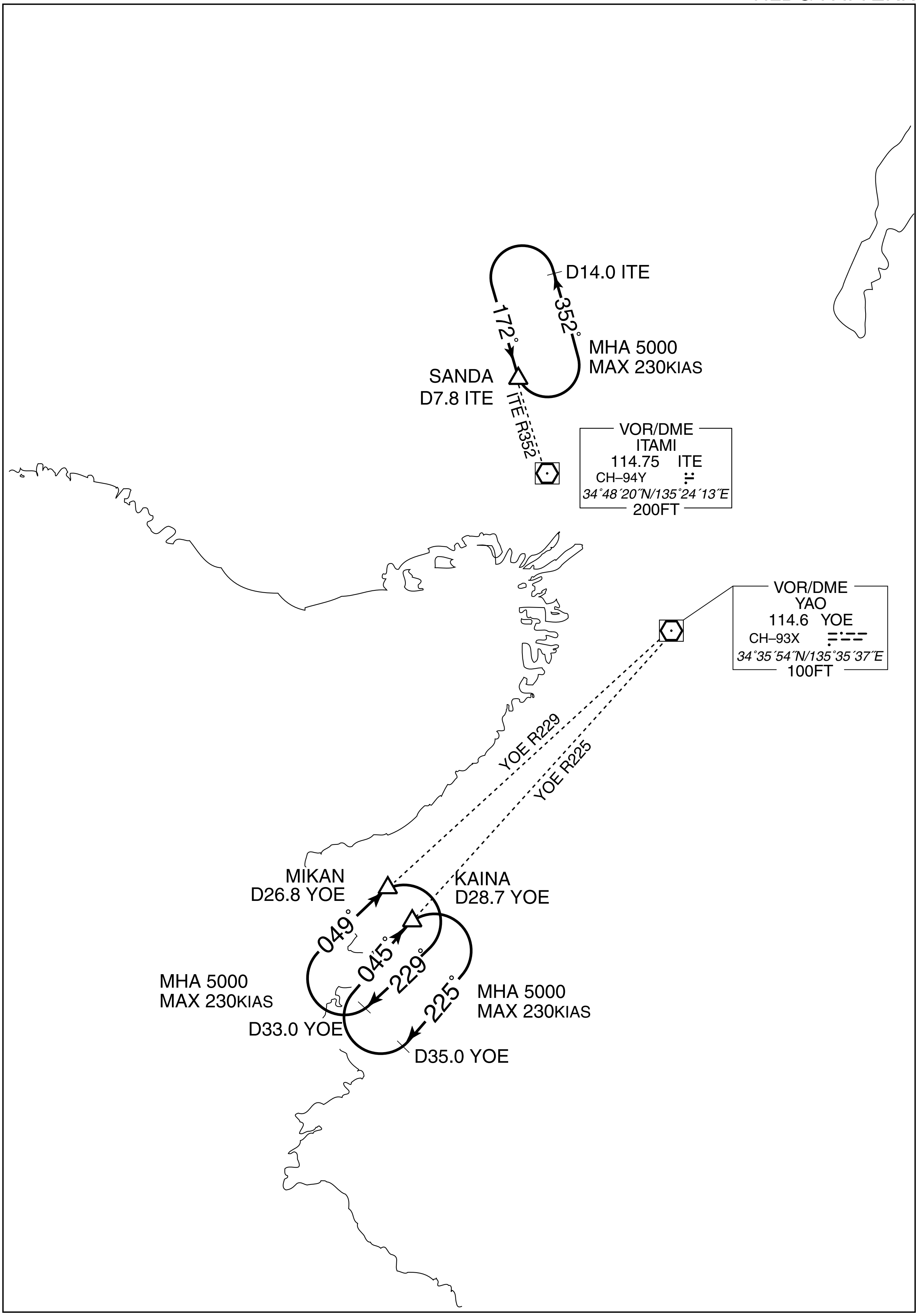
Required additional data

LTP/FTP orthometric height	10.6

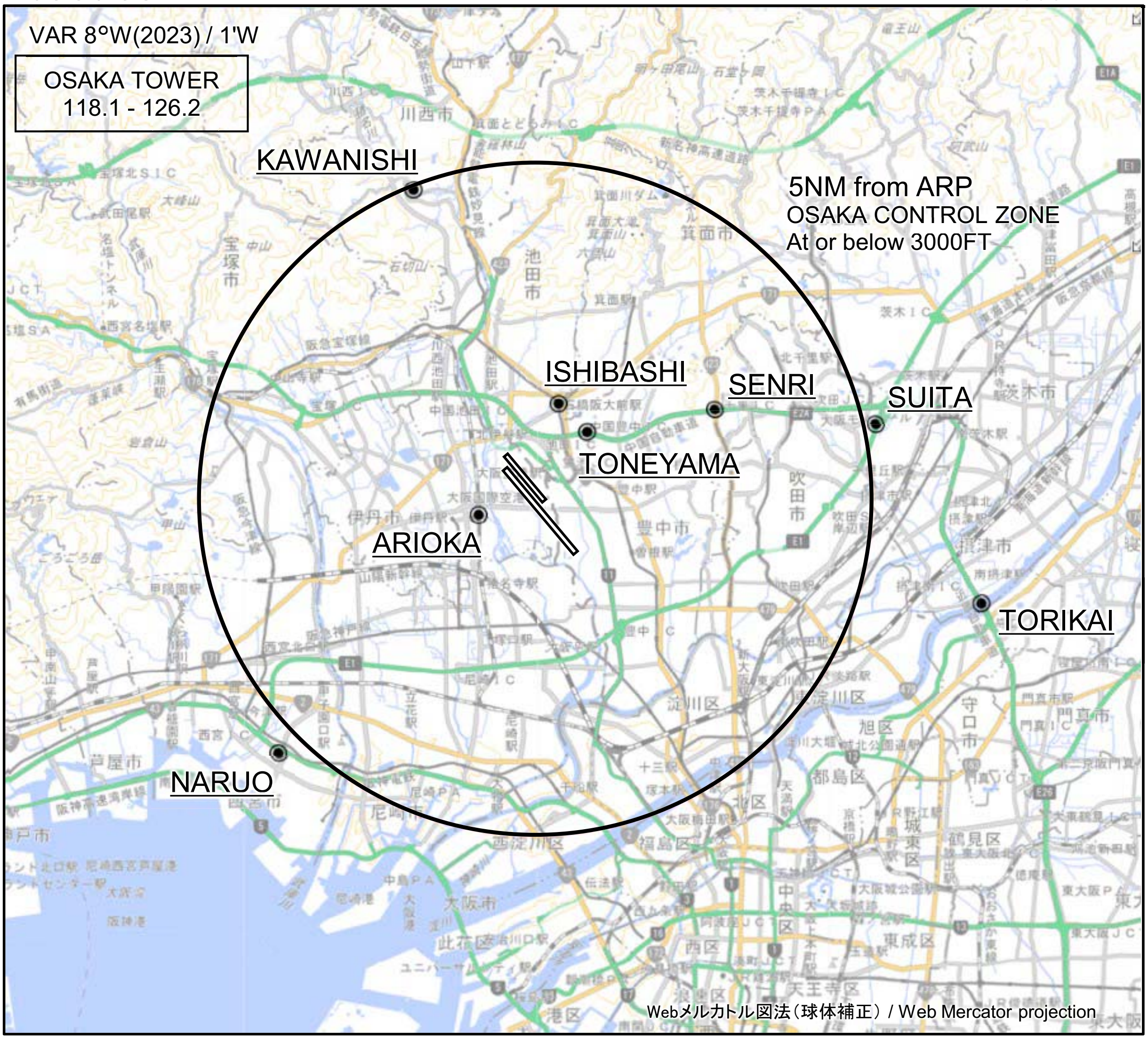
CHANGE : FAS DATA BLOCK, Required additional data established.

RJOO / OSAKA INTL

HLDG PATTERN



RJOO / OSAKA INTL Visual REP



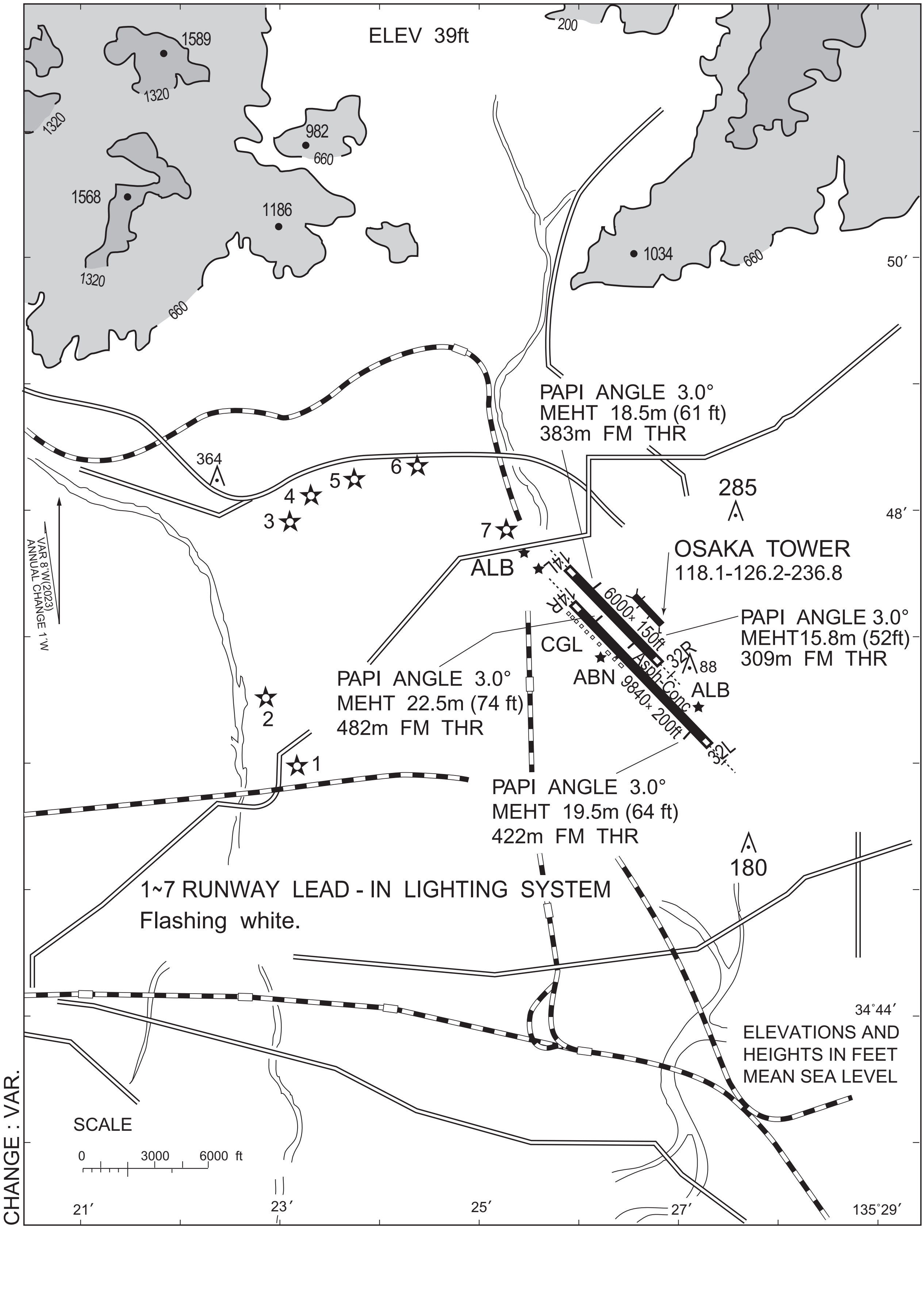
※図中に標高を示す数字がある場合、単位はメートル(m)である。The unit of measurement used to express elevation is meter(m).

Call sign	BRG / DIST from ARP	Remarks
川西 Kawanishi	339°T / 4.9NM	多田神社 Shrine
石橋 Ishibashi	013°T / 1.5NM	阪急石橋阪大前駅 Station
千里 Senri	063°T / 3.0NM	千里インターチェンジ Interchange
吹田 Suita	077°T / 5.2NM	吹田ジャンクション Junction
刀根山 Toneyama	037°T / 1.2NM	中国豊中インターチェンジ Interchange
有岡 Arioka	255°T / 0.9NM	JR伊丹駅 Station
鳥飼 Torikai	103°T / 6.8NM	鳥飼大橋 Bridge
鳴尾 Naruo	225°T / 5.4NM	甲子園球場 Baseball ground

CHANGE : VAR.

RJOO / OSAKA INTL

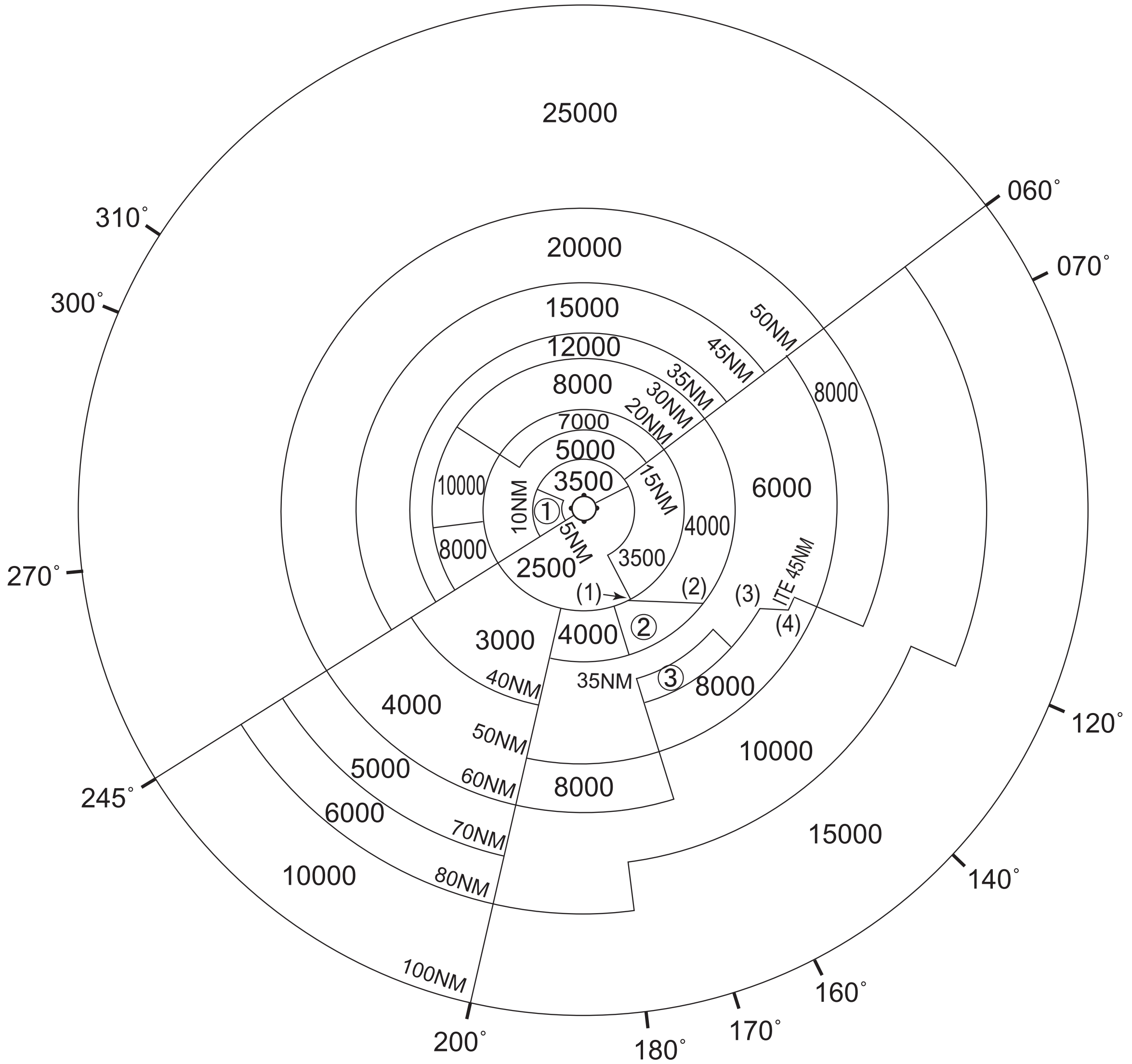
LDG CHART



RJOO / OSAKA INTL

Minimum Vectoring Altitude CHART

VAR 8˚W (2023)



① 4100
② 5000
③ 7000

(1) *342930N/1353527E*
(2) *342925N/1355432E*
(3) *342918N/1360849E*
(4) *342924N/1361335E*

CENTER: *344752N/1352550E* (No.1 RADAR SITE)
CENTER: *344659N/1352600E* (No.2 RADAR SITE)

CHANGE : Update(BTN 245° and 300°).